



KENYATTA UNIVERSITY

DIGITAL SCHOOL OF VIRTUAL AND OPEN LEARNING

IN COLLABORATION WITH

SCHOOL OF PURE & APPLIED SCIENCES

DEPARTMENT: MATHEMATICS AND ACTUARIAL SCIENCE

SMA 112: BASIC MATHEMATICS

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INTRODUCTION

Welcome to this module. The module deals with basic concepts and theorems in Mathematics. This is an interactive instructional module that uses both action and collaborative learning styles that provide you with diverse online learning experiences and effective learning processes. The key purpose of this module is to help you build a good and solid foundation to advanced understanding of mathematical concepts and operations.

In this module, we shall look at counting techniques in mathematics; introduce you to set theory propositional logic, methods of proof and complex numbers.

We hope that you will find this module exciting, educative, and engaging.

BASIC MATHEMATICS FLOW CHART

WEEK	TOPIC
WEEK 1	QUADRATIC EQUATIONS AND INEQUALITIES
WEEK 2	REMAINDER AND FACTOR THEOREM AND THEIR APPLICATIONS
WEEK 3 & 4	PERMUTATIONS AND COMBINATIONS
WEEK 5 & 6	BINOMIAL THEOREM AND APPLICATIONS
WEEK 7 & 8	SET THEORY
WEEK 9	LOGIC
WEEK 10	METHODS OF PROOF
WEEK 11 & 12	COMPLEX NUMBERS
WEEK 13 & 14	EXAMINATION

OVERVIEW OF THE COURSE

Week 0: Introduction (Your Context, Your Goals)

This lesson is intended to help you acclimatize to blended learning and to create a community of learners who will motivate each other during the course. You will be required to introduce yourself to your lecturer and colleagues either physically during a face to face session or even online before other academic interactions start.

Week 1: Quadratic Equations and Inequalities

In this first lesson, we will introduce you to various methods of solving quadratic equations and inequalities. The purpose of this lesson is to help you establish and apply the relationship between roots and coefficients of quadratic equations.

Week 2: Remainder and Factor Theorem and their Applications

In this lesson, we will consider the Remainder and Factor Theorems. These Theorems are useful in solving equations involving polynomials of degree greater than two.

Week 3 & 4: Permutations and Combinations

In this lesson, we introduce you permutations and combinations. Permutations and combinations are very useful tools for counting in Mathematics.

Week 5 & 6: Binomial Theorem and Applications

This lesson considers Binomial Theorem and its applications. This Theorem gives a new way of obtaining coefficients in binomial expansion as opposed to the use of the classical Pascal's Triangle method.

Week 7 & 8: Set Theory

This lesson introduces the abstract concept of set theory and its applications to counting. Algebra of sets will also be considered.

Week 9: Logic

In this lesson you will be introduced to propositional logic. The main emphasis will be determination of truth values of logical statements. Logical statements have truth value either TRUE or FALSE but not both.

Week 10: Methods of Proof

Having studied logic in week 9, we will go a step further in this lesson to consider methods of proof in Mathematics. We will consider both direct and indirect proofs.

Week 11 & 12: Complex Numbers

In weeks 7 and 8, you were introduced to sets. In this lesson we will study one particular set called the set of complex numbers. We will introduce you to arithmetic operations of these numbers, their polar representation and De Moivre's Theorem.

Week 13 & 14: Examination

These two weeks bring together the work you have been doing to an end. This course unit will be examined and will partially contribute to the award of the degree in the programme that you are undertaking.

PURPOSE OF THE MODULE

The purpose of this module is to equip students with basic mathematical skills which build the foundation of Mathematics.

MODULE LEARNING OUTCOMES

By the end of this module the learner will be able to:

- i) Solve quadratic equations and inequalities.
- ii) State and apply the Remainder Theorem and the Factor Theorem
- iii) Differentiate permutations and combinations and apply them in counting iv) Perform set operations and apply set theory to counting
- v) Determine validity of logical arguments

- vi) Prove mathematical statements using direct proof, contradiction, contraposition and induction.
- vii) State and apply De Moivre's Theorem.

COURSE DESCRIPTION

Quadratic equations and inequalities. Remainder and Factor Theorem and their applications. Permutations and combinations, Binomial theorem and its applications. Set theory: Basic operations on sets, **Laws of set theory, Venn diagrams and application**. Logic: Propositions, compound propositions and truth tables. Methods of proof: **Direct, indirect**, Induction, contradiction, cases, **counter examples**. Complex numbers: Arithmetic operations, Geometric representations and polar form. De Moivre's Theorem and its applications.

COURSE REQUIREMENTS

This is a blended learning course that will utilize the flex model. This means that learning materials and instructions will be given online and the lessons will be self-guided with the lecturer being available briefly for face to face sessions and support and also on-site (online) most of the time. You are advised to follow the topic flow-chart given so that you cover at least a lesson every week. You will be required to participate and interact online with your peers and the e-moderator who in this case is your lecturer. Guidelines for the online activities (which we shall keep referring to as e-tivities) will be provided whenever there is an e-tivity. Please note that since the online e-tivities are part of the learning process, they may be graded at the discretion of your e-moderator. Such grading will however be communicated in the e-tivity guidelines and feedback given as soon as possible after the e-tivity. The e-tivities will include but will not be limited to online assessment quizzes, assignments and discussions. There are also assessment questions that you can attempt at the end of every lesson to test your understanding of the lesson. The answers to all the assessment questions are at the end of the module after lesson 8. All the resources that have been used in this module in form of books are available under the resources section after the answers to the questions.

ASSESSMENT

It is important to note that the module has embedded certain learner formative assessment feedback tools that will enable you gauge your own learning progress. The tools include online collaborative discussions forums that focus on team learning and personal mastery and will therefore provide you with peer feedback, lecturer assessment and self- reflection.

I wish you the very best of experiences in this course.

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LESSON ONE**QUADRATIC EQUATIONS AND INEQUALITIES****1.1 Introduction**

In this lesson we discuss quadratic equations and inequalities. In Mathematics we sometimes need to use quadratic equations and inequalities to solve word problems. The most important thing when solving these types of problems is to make sure that they are set up correctly so we can use the quadratic equation and inequalities to easily solve them.

1.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) Solve quadratic equations using factorization, completing square and quadratic formula.
- ii) Solve quadratic inequalities

1.2.1 Quadratic Equations

A quadratic equation is an equation of the form $ax^2 + bx + c = 0$, where a , b and c are real numbers and $a \neq 0$.

1.2.1.1 Methods of solution

- i) Graphical method
- ii) Factorization
- iii) Completing square method
- iv) Quadratic Formula

Example 1.1

Solve $2x^2 - 5x - 3 = 0$ by factorization method.

Solution

Look for two numbers a and b such that $a + b = -5$ and $a \cdot b = -6$, take $a = -6$ and $b = 1$.

Therefore we can write

$$2x^2 - 5x - 3 = 2x^2 - 6x + x - 3 = 0$$

$$\Rightarrow 2x(x - 3) + (x - 3) = 0$$

$$\square (2x + 1)(x - 3) = 0$$

Either $(2x + 1) = 0$ or $(x - 3) = 0$

$$\square x = -\frac{1}{2} \text{ or } x = 3 .$$

NB

$(2x + 1)$ and $(x - 3)$ are called factors of $2x^2 - 5x - 3 = 0$ while $x = -\frac{1}{2}$ and $x = 3$ are called the roots of the equations.

Reverse

$$(x - 3) = 0 \text{ or } (2x + 1) = 0$$

$$\square (x - 3)(2x + 1) = 0$$

$$\square x^2 + \cancel{x} - 3x - \cancel{3} = 0$$

2
2

$$\square 2x^2 + x - 6x - 3 = 0 \text{ or } 2x^2 - 5x - 3 = 0 .$$

Example 1.3

Solve the equation $5x^2 - 6x - 2 = 0$ by completing square method.

Solution

Divide throughout by 5 to get

$$x^2 - \cancel{6}x - \cancel{2} = 0$$

5
5

Or

$$x^2 - \cancel{6}x = 2 \cdot \cancel{5}$$

$\frac{}{5}$
 $\frac{}{5}$

Complete the square by adding $\left(\frac{1}{2} - \frac{6}{5}\right)^2$ on both sides,

$$x^2 - \frac{6}{5}x + \left(\frac{-3}{5}\right)^2 = 2 + \left(\frac{-3}{5}\right)^2 = 2 + \frac{9}{25} = \frac{19}{25}.$$

Factorize the left hand side to get,

$$\left(x - \frac{3}{5}\right)^2 = \frac{19}{25}.$$

Taking the square root on both sides,

$$\frac{3}{5} = \pm \sqrt{\frac{19}{25}} = \pm \frac{\sqrt{19}}{5} \quad x = \quad .$$

$$\frac{3 \pm \sqrt{19}}{5} \quad \pm x =$$

$$\pm x = \frac{3 + \sqrt{19}}{5} \quad \text{or} \quad \pm x = \frac{3 - \sqrt{19}}{5}.$$

Example 1.4

Solve $ax^2 + bx + c = 0$ by completing square method.

Solution

Divide throughout by a to obtain .

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Or

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

$$\pm b \pm$$

Complete the square by adding $\left(\frac{b}{2a}\right)^2$ on both sides,

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} = -\frac{c}{a} + \frac{b^2}{4a^2} = -\frac{c \cdot 4a + b^2}{4a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Factorize the left hand side to get,

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Taking the square root on both sides,

$$\left(x + \frac{b}{2a}\right) = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is called the quadratic formula.

Now, $b^2 - 4ac$ is called the discriminant of the equation.

If $b^2 - 4ac > 0$, then $ax^2 + bx + c = 0$ has two real distinct roots.

If $b^2 - 4ac = 0$, then $ax^2 + bx + c = 0$ has two equal real roots.

If $b^2 - 4ac < 0$, then $ax^2 + bx + c = 0$ has no real roots.

1.2.1.2 Roots of Quadratic Equation

Consider the equation $ax^2 + bx + c = 0$. Then

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \text{i) } a \neq 0$$

Let α and β be roots of equation i), then x

$$= \alpha \text{ or } x = \beta .$$

Therefore,

$$(x - \alpha) = 0 \text{ or } (x - \beta) = 0$$

$$\alpha (x - \beta)(x - \alpha) = 0$$

$$x(x - \beta) - \alpha (x - \alpha) = 0$$

$$\alpha x^2 - \alpha x - \alpha x + \alpha\alpha = 0 \text{ or}$$

Or .

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \quad \text{ii)}$$

Comparing i) and ii), we get

$$-(\alpha + \beta) = -b \quad \alpha\beta = c$$

and

$$\alpha\beta = \frac{c}{a} .$$

Example 1.5

Write down the sums and products of the roots of the following equations;

$$\text{i) } 3x^2 - 2x - 7 = 0$$

$$\text{ii) } 2x(x + 1) = x +$$

$$7 \text{ iii) } 2x^2 + 5x = 1$$

Solution

$$\frac{-b}{a} = \frac{-(-2)}{3} = \frac{2}{3} \text{ and } \Delta = b^2 - 4ac = (-2)^2 - 4(3)(-7) = 4 + 84 = 88$$

$$\text{i) } x = \frac{2 \pm \sqrt{88}}{6} = \frac{2 \pm 2\sqrt{22}}{6} = \frac{1 \pm \sqrt{22}}{3}$$

ii) and iii) Exercise

Example 1.6

Express the following in terms of $x + y$ and xy .

$$\text{i) } x^2 + y^2 \text{ ii) } (x - y)^2$$

$$\text{iii) } x^3 + y^3$$

Solution

i)

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$x^2 + y^2 = (x + y)^2 - 2xy$$

$$\text{ii) } (x - y)^2 = x^2 - 2xy + y^2$$

$$= x^2 + y^2 - 2xy$$

$$= (x + y)^2 - 2xy - 2xy \text{ from part i)}$$

$$= (x + y)^2 - 4xy$$

$$\text{iii) } (x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$= x^3 + y^3 + 3x^2y + 3xy^2$$

$$= x^3 + x^3 + 3xx(x + x)$$

$$xx^3 + x^3 = (x + x)^3 - 3xx(x + x)$$

Example 1.7

The roots of the equation $3x^2 + 4x - 5 = 0$ are α and β . Find the equation whose roots are

- i) $\frac{1}{\alpha^2} - 1$ and $\frac{1}{\beta^2} - 1$
 ii) α and β

Solution

From the given equation,

$$\text{Sum: } \alpha + \beta = \frac{-b}{a} = \frac{-4}{3}$$

$$c \quad -5 \text{ Product: } \alpha\beta = \frac{c}{a} = \frac{-5}{3}$$

$$\frac{-}{a} \quad \frac{-}{3}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{-4}{3}}{\frac{-5}{3}}$$

i) Sum:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{4}{5}$$

and

$$\text{Product: } \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{-3}{-5}$$

$$\frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{3}{5}$$

The required equation is: $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$

$$x^2 - 4x + \frac{3}{5} = 0$$

i.e. x

$$\frac{x^2}{5} - 4x + 3 = 0$$

$$x^2 - 4x - 3 = 0$$

$$i. \text{ or } 5x^2$$

$$- 4x - 3 = 0 .$$

$$ii) \text{ Sum: } x^2 + x^2 = (x + x)^2 - 2xx$$

$$\begin{aligned} &= x^2 - 4x^2 - 5x^2 - 46 \\ &= x^2 - 3x^2 - 2x^2 - 3x^2 = 9 \\ & \quad x^2 \quad x^2 \quad x^2 \quad x^2 \end{aligned}$$

2

$$\text{Product: } x^2 x^2 = (xx)^2 = xx - 5xx = 25$$

$$x^3 x^9$$

$$x^2 - \frac{46}{x} + \frac{9}{25} = 0 \text{ or } 9x^2 - 46x + 25 = 0 . \text{ Required}$$

equation: x

$$9 \quad 9$$

E-tivity 1.2.1: Quadratic Equations

Numbering and pacing and sequencing	1.2.1
Title	Quadratic equations
Purpose	To expose you to the relationship between roots and coefficients of quadratic equations.
Brief summary of overall task	Watch the video on Sum and product of roots of quadratic equations by Mathusay Math Tutorial and then give the formula for the sum and product of roots of quadratic equations in terms of equation's coefficient
Spark	Quadratic Formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Individual contribution	• Watch the video on Sum and product of roots of quadratic equations • Give the formula for the sum and product of roots of quadratic equations in terms of equation's coefficient • Answer the question; 1. Given the roots of the equation $3x^2 + 4x - 5 = 0$ are α and β to find the equation whose roots are; i) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ ii) $(\alpha - 2)$ and $(\beta - 2)$
Interaction begins	• Post your answers on the discussion forum 1.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Quadratic inequalities

1.2.2 Quadratic Inequalities

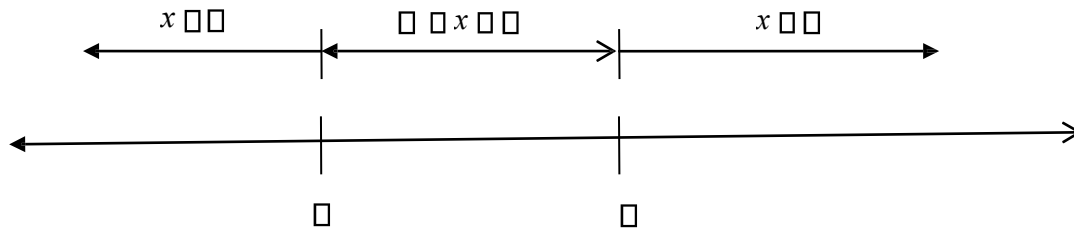
A quadratic inequality is an inequality in which one side is a quadratic polynomial and the other side is zero.

Eg. $2x^2 - x - 6 \geq 0$

Consider the quadratic equation $ax^2 + bx + c = 0$. Now, if $b^2 - 4ac \geq 0$, then this equation has two distinct real roots. Let the roots be α and β with $\alpha \leq \beta$. Then this equation can be written as

$$ax^2 + bx + c = a(x - \alpha)(x - \beta). \quad (1)$$

Next consider the number line below;



Then values of the factors in equation (1) in the three regions on the number line above are as shown in the table below.

	$x - \alpha$	$x - \beta$	$(x - \alpha)(x - \beta)$
$x < \alpha$	Negative	Negative	Positive
$\alpha < x < \beta$	Positive	Negative	Negative
$\beta < x$	Positive	Positive	Positive

When $a > 0$, $ax^2 + bx + c > 0$ when $x < \alpha$ or when $\beta < x$.

$$ax^2 + bx + c < 0 \text{ when } \alpha < x < \beta.$$

When $a < 0$, $ax^2 + bx + c > 0$ when $\alpha < x < \beta$.

$$ax^2 + bx + c < 0 \text{ when } x < \alpha \text{ or when } \beta < x.$$

Example 1.8

Find the set of all values of x for which $2x^2 + 5x - 3 > 0$.

Solution

The roots of the equation $2x^2 + 5x - 3 = 0$ are $\frac{1}{2}$ and -3 . Therefore

$$2x^2 + 5x - 3 = 2(x + 3)(x - \frac{1}{2})$$

and $a = 2 > 0$, $b = -3$ and $c = \frac{1}{2}$. Thus we have

	$x + 3$	$x - \frac{1}{2}$	$2(x + 3)(x - \frac{1}{2})$
$x < -3$	Negative	Negative	Positive
$-3 < x < \frac{1}{2}$	Positive	Negative	Negative
$\frac{1}{2} < x$	Positive	Positive	Positive

Thus $2(x + 3)(x - \frac{1}{2}) > 0$ when $x < -3$ or when $x > \frac{1}{2}$. That is $2x^2 + 5x - 3 > 0$ when $x < -3$ or when $x > \frac{1}{2}$.

Example 1.9

Find the set of all values of x for which $1 + 2x - 3x^2 > 0$.

Solution

The roots of the equation $1 + 2x - 3x^2 = 0$ are $-\frac{1}{3}$ and 1. Therefore

$1 + 2x - 3x^2 = -3(x + \frac{1}{3})(x - 1)$ and we have the table below;

	$x + \frac{1}{3}$	$x - 1$	$(x + \frac{1}{3})(x - 1)$	$-3(x + \frac{1}{3})(x - 1)$
$x < -\frac{1}{3}$	Negative	Negative	Positive	Negative
$-\frac{1}{3} < x < 1$	Positive	Negative	Negative	Positive
$1 < x$	Positive	Positive	Positive	Negative

Thus $-3(x + \frac{1}{3})(x - 1) > 0$ when $x < -\frac{1}{3}$ or $x < 1$. That is $1 + 2x - 3x^2 > 0$ when $x < -\frac{1}{3}$ or $x < 1$.

1.2.2.1 Alternative Method**Steps**

1. Turn the inequality into an equation
2. Find the solutions to the equation
3. Make a number line and check each interval.

Example 1.10

Solve the inequality $x^2 + 2x - 8 \geq 0$.

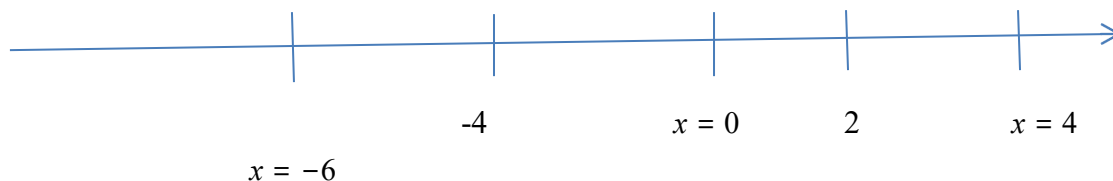
Solution

Solve the equation x^2

$$+ 2x - 8 = 0$$

$$(x + 4)(x - 2) = 0$$

$$\Rightarrow x = -4 \text{ or } x = 2$$



Consider any point less than -4 , say $x = -6$ then

$$(x + 4)(x - 2) = (-6 + 4)(-6 - 2) = (-2)(-8) = 16 \geq 0$$

$\Rightarrow$ the set of all x such that $x \leq -4$ is a solution.

Next, consider any point in the interval $-4 \leq x \leq 2$, say $x = 0$ then

$$(x + 4)(x - 2) = (0 + 4)(0 - 2) = (4)(-2) = -8 \not\geq 0$$

$\Rightarrow$ the set of all x such that $-4 \leq x \leq 2$ is not a solution.

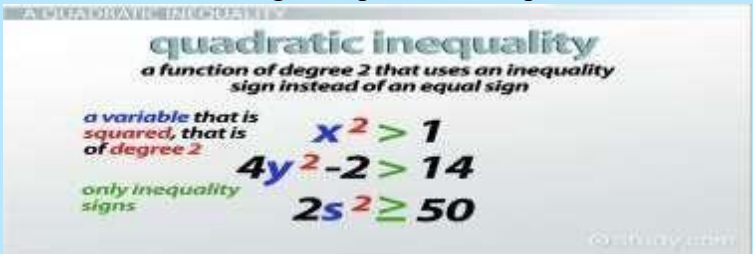
Finally, consider any point greater than 2, say $x = 4$ then

$$(x + 4)(x - 2) = (4 + 4)(4 - 2) = (8)(2) = 16 \geq 0$$

□ the set of all x such that $x \geq 2$ is a solution

Hence the solution to the inequality $x^2 + 2x - 8 \geq 0$ is the set of all x such that either $x \leq -4$ or $x \geq 2$.

E-tivity 1.2.2: Quadratic inequalities

Numbering and pacing and sequencing	1.2.2
Title	Quadratic inequalities
Purpose	To enable the learner to solve quadratic inequalities.
Brief summary of overall task	Watch the video on solving quadratic inequalities by Jeff Suzuki and solve the given quadratic inequalities.
Spark	
Individual contribution	• Watch the video on solving quadratic inequalities. • Solve the following inequalities i) $x^2 - 5x + 6 \geq 0$ ii) $-x^2 + 3x - 2 \geq 0$ iii) $x^2 - 3x \geq 10$
Interaction begins	• Post your answers on discussion forum 1.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind

E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Remainder Theorem

1.3 Assessment

1. The roots of $2x^2 + 3x - 4 = 0$ are α and β . Find the values of

a) $\alpha^2 + \beta^2$ b) $\frac{1}{\alpha} + \frac{1}{\beta}$ c) $(\alpha + 1)(\beta + 1)$

α β

2. Prove that if one root of $ax^2 + bx + c = 0$ is twice the other, then $2b^2 = 9ac$.
3. The sum of the squares of the roots of $ax^2 + bx + c = 0$ is 1, prove that $b^2 = 2ac + a^2$.
4. Find the set of all values of x for which

a) $6 + x \leq 12x^2$

b) $x^2 - 2x - 15 \leq 0$

1.4 References

1. Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
2. Bostock, L., Chandler, S. and Rourke, C. (1982). Further Pure Mathematics. Stanley Thornes Publishers Ltd.
3. Larson, R. and Hostetler, R. (1997). Precalculus, Houghton Company.

LESSON TWO

REMAINDER AND FACTOR THEOREM AND THEIR APPLICATIONS

2.1 Introduction

In this lesson we will study the Remainder theorem and Factor Theorem and their applications. The applications include factorization of polynomials of degree greater than two and solution of equations involving polynomials of degree 3 and higher.

2.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) State and apply the Remainder Theorem ii) State and apply the Factor Theorem

2.2.1 The Remainder Theorem

2.2.1.1 Long Division of Polynomials

A polynomial in x of degree n is an expression of the form

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0, \text{ where } a_0, a_1, a_2, \dots, a_n \text{ are real numbers and } a_n \neq 0.$$

Eg $f(x) = 2x^3 - 4x^2 + x + 1$ is a polynomial of degree 3.

When a polynomial $f(x)$ is divided by another polynomial $g(x)$, we obtain a quotient $q(x)$ and a remainder $r(x)$. This is similar to division of two integers in arithmetic operations.

The degree of the remainder $r(x)$ is less than that of the divisor $g(x)$.

$$\begin{array}{c} \frac{(\quad)}{g(x)} \overline{f(x)} \\ \dots \\ \dots \quad q x \\ r(x) \end{array}$$

We write $f(x) = g(x)q(x) + r(x)$ or $\underline{f(x)} = q(x) + \underline{r(x)}$.

$$g(x) \qquad g(x)$$

Example 2.1

Divide a) $f(x) = 3x^3 + 2x^2 + x - 1$ by i) $g(x) = x - 1$

ii) $g(x) = x^2 - 2$

b) $f(x) = 4x^3 - x + 2$ by $g(x) = 3x + 2$

Solution

$$\begin{array}{r}
 \overline{3x^3 + 2x^2 + x - 1} \\
 \underline{3x^3 - 3x^2} \\
 5x^2 + x \\
 \underline{5x^2 - 5x} \\
 6x - 1 \\
 \underline{6x - 6} \\
 5
 \end{array}$$

□ $q(x) = 3x^2 + 5x + 6$ and $r(x) = 5$

$$\begin{array}{r}
 \overline{3x^3 + 2x^2 + x - 1} \\
 \underline{3x^3 - 6x} \\
 2x^2 + 7x - 1 \\
 \underline{2x^2 - 4} \\
 7x + 3
 \end{array}$$

$$\square q(x) = 3x + 2 \text{ and } r(x) = 7x + 3$$

$$\begin{array}{r} \text{b) } 3x + \frac{4}{3}x^2 - \frac{8}{9}x + \frac{7}{27} \quad 2 \\ \hline 4x^3 - x + 2 \\ \underline{4x^3 + \frac{8}{3}x^2} \\ -\frac{8}{3}x^2 - x \\ \underline{-\frac{8}{3}x^2 - \frac{16}{9}x} \\ \frac{7}{9}x + 2 \\ \underline{\frac{7}{9}x + \frac{14}{27}} \\ \frac{40}{27} \end{array}$$

$$\left(\frac{4}{3}x^2 - \frac{8}{9}x + \frac{7}{27} \right) \quad \text{Therefore } q(x) = \quad \text{and } r(x) = \frac{40}{27}.$$

Theorem 2.1 (The Remainder Theorem)

If a polynomial $f(x)$ is divided by $x - a$, the remainder is $f(a)$.

Proof

Suppose that when $f(x)$ is divided by $x - a$ the remainder is $r(x)$. Then by division algorithm,

$$f(x) = (x - a)q(x) + r(x).$$

Putting $x = a$, we get $f(a)$

$$= 0 \cdot q(a) + r(a)$$

$$\square \text{ Remainder} = r(a) = f(a)$$

□

Example 2.2

Find the remainder when $x^5 - 4x^3 + 2x + 3$ is divided by a) $x - 1$

b) $x + 2$

Solution

Let $f(x) = x^5 - 4x^3 + 2x + 3$, then

a) The remainder when $f(x)$ is divided by $x - 1$ is

$$f(1) = 1 - 4 + 2 + 3 = 2$$

b) The remainder when $f(x)$ is divided by $x + 2$ is

$$f(-2) = (-2)^5 - 4(-2)^3 + 2(-2) + 3 = -1$$

Example 2.3

Prove that when a polynomial $f(x)$ is divided by $ax + b$, where $a \neq 0$, the remainder is $f\left(-\frac{b}{a}\right)$. Hence find the remainder when $4x^3 - 6x + 5$ is divided by $(2x - 1)$.

✎

□ □

Proof

Suppose that when $f(x)$ is divided by $ax + b$, where $a \neq 0$ the remainder is $r(x)$. Then

$$f(x) = (ax + b)q(x) + r(x).$$

$\frac{b}{a}$

Putting $x = -\frac{b}{a}$, we
get

$$\begin{aligned} f\left(-\frac{b}{a}\right) &= 0 \cdot q\left(-\frac{b}{a}\right) + r\left(-\frac{b}{a}\right) \\ &= r\left(-\frac{b}{a}\right) \end{aligned}$$

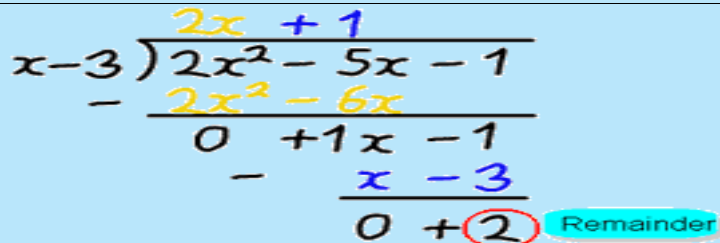
$$\therefore \text{Remainder} = r\left(-\frac{b}{a}\right)$$

$$= f\left(-\frac{b}{a}\right) \quad \square$$

$$f\left(\frac{-1}{2}\right) = f\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right)^3 - 6\left(\frac{1}{2}\right) + 5 = \frac{5}{2}.$$

If $f(x) = 4x^3 - 6x + 5$ is divided by $2x - 1$, the remainder is

E-tivity 2.2.1: The Remainder Theorem

Numbering and pacing and sequencing	2.2.1
Title	The Remainder Theorem
Purpose	To help you to find the remainder when a polynomial $f(x)$ is divided by a linear factor of the form $(x - a)$
Brief summary of overall task	Watch the video on Remainder Theorem by Jeff Suzuki
	by and solve the given questions.
Spark	
Individual contribution	- Watch the video on Remainder Theorem. - Find the remainder when i) $x^3 - 2x + 4$ is divided by $x - 1$ ii) $2x^3 - x^2 + 2$ is divided by $x - 3$
Interaction begins	- Post your answers on discussion forum 2.2.1 - Read what your colleagues have posted. - In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	- Focussing group discussion - Encouraging lurkers (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This activity should take one hour.
Next	Factor Theorem

2.2.2 The Factor Theorem**Cororally 2.2 (The Factor Theorem)**

If for a given function $f(x)$, $f(a) = 0$ then $(x - a)$ is a factor of $f(x)$.

That is if $(x - a)$ is a factor of $f(x)$, then there will be no remainder when $f(x)$ is divided by $(x - a)$;

$$R = 0 \iff f(a) = 0.$$

Example 2.4

Factorize $x^4 - 3x^3 + 4x^2 - 8$ completely.

Solution

We use try and error method together with the factor theorem.

Let $f(x) = x^4 - 3x^3 + 4x^2 - 8$, then

$$f(1) = 1 - 3 + 4 - 8 \neq 0 \implies x - 1 \text{ is not a factor of } f(x)$$

$$f(2) = 16 - 24 + 16 - 8 = 0 \implies x - 2 \text{ is a factor of } f(x)$$

Dividing,

$$\begin{array}{r} x^3 - x^2 + 2x + 8 \\ 4x - 8 \overline{) x^4 - 3x^3 + 4x^2 - 8} \end{array}$$

$$-x^3 + 4x^2$$

$$\underline{-x^3 + 2x^2}$$

$$\begin{array}{r}
 2x^2 - 8 \\
 \underline{2x^2 - 4x} \\
 4x - 8 \\
 \underline{4x - 8} \\
 \dots
 \end{array}$$

Next, we factorize $f_1(x) = x^3 - x^2 + 2x + 4$. Now

$$f_1(1) = 1 - 1 + 2 + 4 \neq 0 \Rightarrow x - 1 \text{ is not a factor of } f_1(x)$$

$$f_1(2) = 8 - 4 + 4 + 4 \neq 0 \Rightarrow x - 2 \text{ is not a factor of } f_1(x)$$

$$f_1(-1) = -1 - 1 - 2 + 4 \neq 0 \Rightarrow x + 1 \text{ is a factor of } f_1(x)$$

$$\begin{array}{r}
 x^2 - 2x + 4 \\
 x + 1 \overline{) x^3 - x^2 + 2x + 4}
 \end{array}$$

$$\begin{array}{r}
 x^3 + x^2 \\
 - 2x^2 + 2x \\
 \underline{-2x^2 - 2x}
 \end{array}$$

$$\begin{array}{r}
 4x + 4 \\
 \underline{4x + 4} \\
 \dots
 \end{array}$$

By using try and error method, it can be shown that $x^2 - 2x + 4$ no linear factors. Hence x^4

$$- 3x^3 + 4x^2 - 8 = (x - 2)(x + 1)(x^2 - 2x + 4)$$

Example 2.5

Solve the equation $x^3 - 7x - 6 = 0$.

Solution

Let $f(x) = x^3 - 7x - 6$ then,

$$f(1) = 1 - 7 - 6 \neq 0 \quad \square \quad x - 1 \text{ is not a factor of } f(x)$$

$$f(2) = 8 - 14 - 6 \neq 0 \quad \square \quad x - 2 \text{ is not a factor of } f(x)$$

$$f(-1) = -1 + 7 - 6 = 0 \quad \square \quad x + 1 \text{ is a factor of } f(x)$$

$$\begin{array}{r} x^3 - x - 6 \quad | \quad x^3 - 7x \\ \underline{-x^3 + x^2} \\ -x^2 - 7x \\ \underline{-x^2 - x} \\ -6x - 6 \\ \underline{-6x - 6} \\ 0 \end{array}$$

$$x^3 - x - 6 = (x + 1)(x - 3)(x + 2)$$

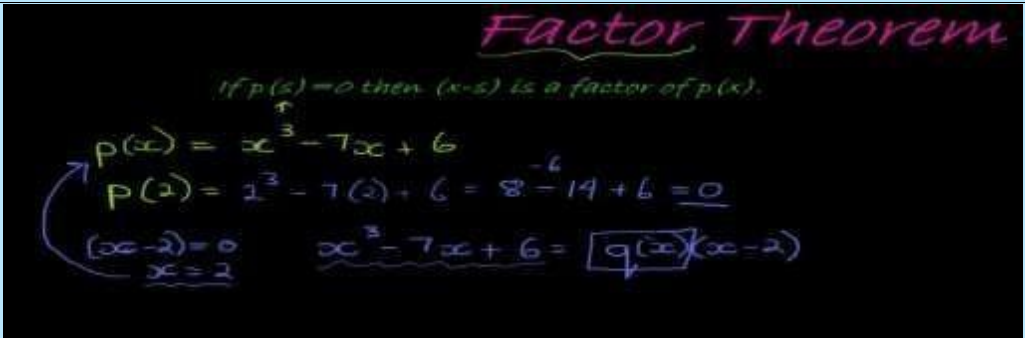
Therefore

$$(x + 1)(x - 3)(x + 2) = 0$$

$$\square \quad x = -1, \text{ or } x = 3 \text{ or } x = -2.$$

Activity 2.2.2: The Factor Theorem

Numbering and pacing and sequencing	2.2.2
Title	The Factor Theorem

Purpose	To help you to factorize and solve equations involving polynomials of degree ≤ 3
Brief summary of overall task	Watch the video on Factor Theorem by Elroi Academy and solve the given questions.
Spark	
Individual contribution	• Watch the video on Factor Theorem. • Factorize $x^3 - 6x^2 + 11x - 6$ • Solve the equation $3x^3 + x^2 - 5x + 2 = 0$
Interaction begins	• Post your answers on discussion forum 2.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Permutations

2.3 Assessment

- Find the remainder when $2x^3 - x^2 + 3x - 5$ is divided by
 - $2x+1$
 - $3x-2$
- A cubic polynomial $ax^3 + bx + 6$ is divisible by $x+2$. It leaves a remainder -3 when divided by $x-1$. Determine the values of a and b and factorize the polynomial completely.

2.4 References

- Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
- Bostock, L., Chandler, S. and Rourke, C. (1982). Further Pure Mathematics. Stanley Thornes Publishers Ltd.

3. Larson, R. and Hostetler, R. (1997). Precalculus, Houghton Company.

LESSON THREE

PERMUTATIONS AND COMBINATIONS

3.1 Introduction

In this lesson we will study Permutations and Combinations. In mathematics, the notion of permutation relates to the act of permuting, or rearranging, members of a set into a particular sequence or order. Permutations occur, in more or less prominent ways, in almost every area of mathematics. They often arise when different orderings on certain finite sets are considered, possibly only because one wants to ignore such orderings and needs to know how many configurations are thus identified. On the other hand combinations are selections that disregard order.

3.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) Apply permutations in counting
- ii) Apply combinations in counting

3.2.1 Permutations

3.2.1.1 The Fundamental Counting Principle

The Fundamental Counting Principle gives the guiding rule for finding the number of ways to accomplish two tasks.

Theorem 3.1 [The Fundamental Counting Principle]

If there are m ways to do one thing, and n ways to do another, then there are $m \times n$ ways of doing both.

Example 3.1

- a) Let's say you want to flip a coin and roll a die. There are 2 ways that you can flip a coin and 6 ways that you can roll a die. Therefore there are $2 \times 6 = 12$ ways that you can flip a coin and roll a die.

- b) If you want to draw 2 cards from a standard of 52 cards without replacing them, then there are 52 ways to draw the first and 51 ways to draw the second, so there are a total of $52 \times 51 = 2652$ ways to draw the two cards.

Definition 3.1 (Permutations)

A permutation is an arrangement of a number of objects in a particular order.

Example 3.2

In how many ways can letters A, B and C be arranged in three consecutive positions?

Solution

ABC ☐

☐

ACB ☐

BAC ☐

☐ Six permutations BCA ☐

CAB ☐

☐

CBA ☐

Or equivalently,

1 st	2 nd	3 rd
A	B	C
	C	B
B	A	C
	C	A
C	A	B
	B	A

That is

1. If we have ABC, we can choose either A or B or C for the first position. Therefore there are 3 distinct ways of filling the first position.
2. Once the first position is filled two letters remain and any of them can be used in the 2nd position. For each of the 3 ways of filling the 1st position, we have two ways of filling the second position.

3. The third position can be filled in any of one way.

Therefore total number of arrangements is

$$3 \times 2 \times 1 = 6 \text{ ways.}$$

Similarly, if there are four objects, then there are

$$4 \times 3 \times 2 \times 1 = 24 \text{ ways of arranging them and so on}$$

Each of the arrangements in the above example is called a permutation.

Definition 3.2

The number of ways of arranging n distinct objects is n

$$(n-1)(n-2) \dots 3 \times 2 \times 1$$

which is denoted by $n!$ and read as ' n factorial.'

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$3! = 3 \times 2 \times 1 = 6$$

$$2! = 2 \times 1 = 2$$

$$1! = 1$$

By convention $0! = 1$, the number of ways of arranging zero objects.

Example 3.3

Evaluate

$$\text{a) } \frac{9!}{2! \cdot 7!} \quad \frac{6!}{(3!)^2}$$

Solution

$$\text{a) } \frac{9!}{2! \cdot 7!} = \frac{9 \times 8 \times 7!}{2! \cdot 7!} = \frac{9 \times 8}{2} = 36$$

$$\text{b) } \frac{6!}{(3!)^2} = \frac{6 \times 5 \times 4 \times 3!}{3! \cdot 3!} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$$

Example 3.4

Write $40 \times 39 \times 38 \times 37$ in factorial notation.

Solution

$$\begin{aligned}
 40 \times 39 \times 38 \times 37 &= 40 \times 39 \times 38 \times 37 \times \frac{36 \times 35 \times \dots \times 3 \times 2 \times 1}{36 \times 35 \times \dots \times 3 \times 2 \times 1} \\
 &= \frac{40!}{36!}
 \end{aligned}$$

Now, consider a situation where 4 letters are to be filled in two positions. How many arrangements or permutations are there for 2 objects chosen from 4 objects?

- i) There are 4 ways of filling the first position
- ii) Once the first position is filled, three letters remain; therefore there are 3 ways of filling the second position.

Therefore the total number of arrangements of 2 objects chosen from 4 objects will be

$$4 \times 3 = 12.$$

$$\text{But } 4 \times 3 = \frac{4!}{(4-2)!}.$$

In the same way,

$5 \times 4 = 20$ is the number of arrangements of 2 objects chosen from 5 objects.

$$\text{But } 5 \times 4 = \frac{5!}{3!(5-2)!}.$$

$5 \times 4 \times 3 = 60$ is the number of arrangements of three objects chosen from 5 distinct object.

$$\text{But } 5 \times 4 \times 3 = \frac{5!}{2!(5-3)!}.$$

Definition 3.3

The number of arrangements or permutations of r objects chosen from n distinct objects denoted by nP_r is given by

$${}^nP_r = \frac{n!}{(n-r)!}, \quad r \leq n \quad \text{and } n \text{ is an integer.}$$

Example 3.5

In a lottery a total of a thousand tickets were sold. Determine the winners of the 1st, 2nd, and the 3rd prizes if three tickets are drawn one after the other?

Solution

This is a permutation of 3 tickets chosen from 1000 tickets. Therefore we have,

$$\begin{aligned} {}^{1000}P_3 &= \frac{1000!}{(1000-3)!} = \frac{1000!}{997!} \\ &= \frac{1000 \times 999 \times 998 \times 997!}{997!} \\ &= 9.97 \times 10^8 \text{ ways.} \end{aligned}$$

Example 3.6

Five letter words are formed from letters of the word TIEGROUP. How many of these have no repeated words?

Solution

These are distinct arrangements of 5 letters chosen from 8 letters. Thus we get

$$\begin{aligned} {}^8P_5 &= \frac{8!}{(8-5)!} = \frac{8!}{3!} \\ &= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3!}{3!} \end{aligned}$$

$$=6720.$$

Remark 3.1

In some cases the objects to be arranged may not be distinct as some may be repeated. **E.g.**

BOB, BABA, CHOOSE etc.

If we have n objects of which r_1 of the first kind are alike, r_2 of the second kind are alike, ..., r_k of the k^{th} kind are alike, then the number of distinct permutations is

$$\frac{n!}{r_1! r_2! \cdots r_k!}.$$

Example 3.7

Determine the number of permutations of letters of the words

a) ARRANGE

b) SPEEDOMETERS

c) MISSISSIPPI

Solution

$$\text{i) } \frac{7!}{2! 4!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1 \times 2 \times 1} = 105$$

$$\text{ii) } \frac{12!}{9! 2!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 2 \times 1} = 66$$

$$\text{iii) } \frac{11!}{4! 4! 2!} = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1 \times 2 \times 1} = 165$$

Example 3.8

How many even numbers, greater than 2000, can be formed with the digits 1, 2, 4, 8 if each digit can be used only once in each number?

Solution

If the number is greater than 2000, the first digit can be chosen in 3 ways i.e. 2, 4 and 8. Then whichever has been chosen to be the first digit, there are two ways in which the last digit may be chosen, in order to make the number even. The second digit can be chosen in two ways and this leaves one integer for the third place.

Therefore there are $3 \times 2 \times 1 \times 2 = 12$ ways.

1 st	2 nd	3 rd	4 th
3	2	1	2

Example 3.9

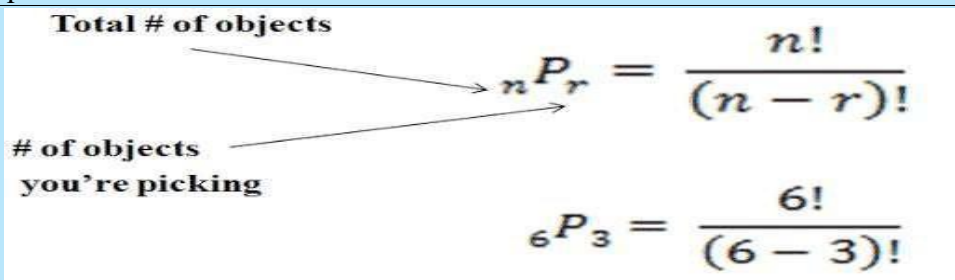
How many arrangements can be made of the letters chosen from PEAT if the first letter is a vowel and each arrangement contains three different letters?

Solution

1 st	2 nd	3 rd
2	3	2

Total arrangements = $2 \times 3 \times 2 = 12$.

Etivity 3.2.1: Permutations

Numbering and pacing and sequencing	3.2.1
Title	Permutations
Purpose	To help you solve counting problems using permutations.
Brief summary of overall task	Watch the video on Permutations by OCL Phase 2 and solve the given questions.
Spark	 The diagram illustrates the permutation formula ${}_nP_r = \frac{n!}{(n-r)!}$. It shows two arrows: one from 'Total # of objects' to 'n' and another from '# of objects you're picking' to 'r'. Below the formula, an example is given: ${}_6P_3 = \frac{6!}{(6-3)!}$.

Individual contribution	• Watch the video on permutations. • Answer the following questions; 1. Express $12 \times 11 \times 10 \times 9$ in factorial notation. 2. How many 4 letters words can be formed using letters of the word BROWSER? 3. In how many ways can five different books be arranged on a shelf? 4. In how many ways can eight people be seated at a round table?
Interaction begins	• Post your answers on discussion forum 3.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Combinations

3.2.2 Combinations

In permutations, the order in which objects are chosen is important. In some cases, the order of selection is irrelevant. When the selection of objects is made with no regard being paid to order, it is referred to as a combination.

Thus ABC, ACB, CBA are different permutations but they are the same combinations of letters.

Example 3.10

How many committees of three persons can be made of five persons namely; Anderson, Beatrice, Charles, Duncan and Eunice?

Solution

Let A Andrew	ABC	BCD	CDE
B Beatrice	ABD	BCE	
C Charles	ABE	BDE	
D Duncan	ACD		
E Eunice	ACE		

ADE

Therefore there are 10 different committees or combinations of 3 persons chosen from five. However there are ${}^5P_3 = 60$ distinct permutations of 3 letters chosen from the letters of the word ABCDE. Thus one combination of three letters can be rearranged to give $3! = 6$ permutations.

Definition 3.4

In general, the number of combinations of r objects chosen from n distinct objects is given by

$${}_nC_r = \frac{n!}{r!(n-r)!} \text{ or } {}_nC_r = \frac{n!}{r! \cdot (n-r)!}$$

In the previous example, the number of combinations of 3 persons chosen from 5 is

$${}_5C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1} = \frac{5 \times 4}{2} = 10$$

Example 3.11

Five policemen are to be selected for duty from a force of 20. In how many ways can this be done?

Solution

$${}_{20}C_5 = \frac{20!}{(20-5)!5!} = \frac{20!}{5!15!} = 15,504.$$

Example 3.12

A mixed hockey team containing 5 men and 6 women is to be chosen from 7 men and 9 women. In how many ways can this be done?

Solution

5 men can be chosen from 7 men in 7C_5 ways and 6 women can be chosen from 9 women in 9C_6 ways. Therefore by the Fundamental Counting Principle, the total number of ways of selecting the team is

$$\begin{aligned}
 {}^7C_5 \square {}^9C_6 &= \frac{7!}{5!2!} \square \frac{9!}{6!3!} \\
 &= \frac{5!(7-5)!}{7!} \square \frac{6!(9-6)!}{9!} \\
 &= \frac{5!2!}{7!} \square \frac{6!3!}{9!} \\
 &= 21 \square 84 \\
 &= 1764
 \end{aligned}$$

Example 3.13

A girl wants to invite 8 friends but there is only room for 4 of them. In how many ways can she choose whom to invite if two of them are sisters and must not be separated?

Solution

Consider two cases

i) When both sisters are invited ii)

When both sisters are not invited

From i), if the two sisters are to be invited, we select 2 out of 6 in

$${}^6C_2 = \frac{6!}{2!4!} = 15 \text{ ways.}$$

From ii), if the two sisters are not invited, we select 4 out of 6 in

$${}^6C_4 = \frac{6!}{4!2!} = 15 \text{ ways.}$$

Therefore the total number of ways is 15

+15 = 30.

E-tivity 3.2.2: Combinations

Numbering and pacing and sequencing	3.2.2
Title	Combinations
Purpose	To help you to count using combinations.
Brief summary of overall task Spark	Watch the video on Combinations by OCL Phase 2 and answer the given questions. Combination Formula A combination is a grouping or subset of items. For a combination, the order does not matter. $C(n, r) = {}^nC_r = \frac{n!}{(n-r)!r!}$ Number of items in set Number of items selected from the set
Individual contribution	• Watch the video on combinations. • Answer the following questions; 1. A committee of ten is to be formed from 9 men and 6 women. In how many ways can the members be chosen if at least 4 women are to be in the committee? 2. How many different hands of five cards can be dealt from a suit of 13 cards?
Interaction begins	• Post your answers on discussion forum 3.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion

Schedule and time	This activity should take one hour.
Next	Binomial Theorem for a positive integer

3.3 Assessment

1. Simplify $\frac{15!}{11! 4!} + \frac{15!}{12! 3!}$
2. How many odd numbers, greater than 500, 000 can be made from the digits 2, 3, 4, 5, 6, 7
 - a) Without repetition
 - b) with repetition allowed?
3. A committee of six is to be formed from 9 women and 3 men. In how many ways can the members be chosen so as to include at most 2 men?

3.4 References

1. Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
2. Bostock, L., Chandler, S. and Rourke, C. (1982). Further Pure Mathematics. Stanley Thornes Publishers Ltd.
4. Larson, R. and Hostetler, R. (1997). Precalculus, Houghton Company.

LECTURE FOUR**BINOMIAL THEOREM AND ITS APPLICATIONS****4.1 Introduction**

In this lesson we will study the Binomial Theorem and its applications. The Binomial Theorem describes the algebraic expansion of powers of a binomial without actually multiplying out.

4.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) State and apply the Binomial Theorem for a positive integer.

- ii) . State and apply the Binomial Theorem for any index.

4.2.1 Binomial Theorem for a positive integer

4.2.1.1 Expansion using the Pascal's Triangle

Definition 4.1

A binomial is a sum or a difference of two terms. **E.g.**

$x + y$, $2x + 3y$ are binomials.

Consider the following expansions;

$$(a + b)^0 = 1$$

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Writing the coefficients in a triangular array gives;

$$\begin{array}{cccccc}
 & & & & & 1 \\
 & & & & 1 & & 1 \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1
 \end{array}$$

This triangular array is called the Pascal's Triangle (up to degree 4). The entries of any row in the Pascal's triangle can be obtained from the row immediately above it by simply adding consecutive entries of the latter row.

Example 4.1

Expand $(2x + 3y)^3$ in descending powers of x .

Solution

This is a binomial of degree 3. Thus from the Pascal's triangle the coefficients are

1, 3, 3, 1

Therefore letting $a = 2x$ and $b = 3y$ we have,

$$\begin{aligned}(2x + 3y)^3 &= (2x)^3 + 3(2x)^2(3y) + 3(2x)(3y)^2 + (3y)^3 \\ &= 8x^3 + 36x^2y + 54xy^2 + 27y^3\end{aligned}$$

Example 4.2

$$(2x - \frac{1}{2})^4$$

Obtain the expansion of $(2x - \frac{1}{2})^4$, in descending powers of x .

$$(2x - \frac{1}{2})^4$$

Solution

This is a binomial of degree 4. Thus from the Pascal's triangle the coefficients are

1, 4, 6, 4, 1

Therefore letting $a = 2x$ and $b = -\frac{1}{2}$ we have,

$$\begin{aligned}(2x - \frac{1}{2})^4 &= (2x)^4 + 4(2x)^3(-\frac{1}{2}) + 6(2x)^2(-\frac{1}{2})^2 + 4(2x)(-\frac{1}{2})^3 + (-\frac{1}{2})^4 \\ &= 16x^4 + 4(8x^3)(-\frac{1}{2}) + 6(4x^2)(\frac{1}{4}) + 4(2x)(-\frac{1}{8}) + \frac{1}{16} \\ &= 16x^4 - 16x^3 + 6x^2 - x + \frac{1}{16}\end{aligned}$$

4.2.1.2 Binomial Theorem

Consider the binomial $(a + b)^3$ of degree 3. Expanding we get

$$a^3 + 3a^2b + 3ab^2 + b^3.$$

Observe that the coefficients in this expansion can also be obtained as follows;

$$1 = \frac{3!}{3!0!}, 3 = \frac{3!}{2!1!}, 3 = \frac{3!}{1!2!}, 1 = \frac{3!}{0!3!}.$$

In general, we have

Theorem

4.1 (The Binomial Theorem)

If n is a positive integer,

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \cdots + \binom{n}{n} b^n, \text{ where}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Example 4.3

Find the coefficient of x^{10} in the expansion of $(2x - 3)^{14}$.

Solution

$$(2x - 3)^{14} = \binom{14}{0} (2x)^{14} + \binom{14}{1} (2x)^{13} (-3) + \cdots + \binom{14}{10} (2x)^4 (-3)^{10} + \cdots$$

Therefore the coefficient of x^{10} is

$$\binom{14}{10} (2)^4 (-3)^{10} = 83,026,944.$$

Example 4.4

$$(1+x)^{10}$$

Obtain the first four terms of the expansion of $(1+x)^{10}$ in ascending powers of x . Hence find

$$(1.005)^{10}$$

the value of $(1.005)^{10}$ correct to four decimal places.

Solution

$$\begin{aligned} (1+x)^{10} &= {}^{10}C_0 + {}^{10}C_1 x + {}^{10}C_2 x^2 + {}^{10}C_3 x^3 + \dots \\ &= 1 + 10x + \frac{10 \times 9}{2} x^2 + \frac{10 \times 9 \times 8}{6} x^3 + \dots \\ &= 1 + 10x + 45x^2 + 120x^3 + \dots \end{aligned} \quad (6.1)$$

Since

$$(1.005)^{10} = (1+0.005)^{10}$$

and comparing with $(1+x)^{10}$ we get

$$x = 0.005$$

$$x = 0.005$$

$$2$$

$$2x = 2 \times 0.005 = 0.01$$

Substituting this value of x in equation (6.1) we obtain

$$\begin{aligned} (1.005)^{10} &= 1 + 10(0.005) + \frac{10 \times 9}{2} (0.005)^2 + \frac{10 \times 9 \times 8}{6} (0.005)^3 + \dots \\ &= 1 + 0.05 + 0.001125 + 0.000015 + \dots \\ &= 1.0511 \quad (4 \text{ dp}) \end{aligned}$$

Example 4.5

Obtain the expansion of $(1+x-2x^2)^8$ as far as the term in x^3 .

Solution

$$\begin{aligned}
 (1 + x - 2x^2)^8 &= {}^8P_0 1 + {}^8P_1 (x - 2x^2) + {}^8P_2 (x - 2x^2)^2 + {}^8P_3 (x - 2x^2)^3 + \dots \\
 &= 1 + {}^8P_1 (x - 2x^2) + {}^8P_2 (x^2 - 4x^3 + 4x^4) + {}^8P_3 (x^3 - 6x^4 + 12x^5 - 8x^6) + \dots \\
 &= 1 + 8(x - 2x^2) + 28(x^2 - 4x^3 + 4x^4) + 56(x^3 - 6x^4 + 12x^5 - 8x^6) + \dots \\
 &= 1 + 8x - 16x^2 + 28x^2 - 112x^3 + 56x^3 + \dots \\
 &= 1 + 8x + 12x^2 - 56x^3 + \dots
 \end{aligned}$$

Activity 4.2.1: Binomial Theorem for a positive integer

Numbering and pacing and sequencing	4.2.1
Title	Binomial Theorem for a positive integer
Purpose	To you to carry out expansion using Binomial Theorem for a positive integer
Brief summary of overall task	Watch the video on Binomial Theorem General term by Manoj Singh and answer the given questions.

Spark	Binomial Theorem $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$ www.GIMathS.Com
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Individual contribution	• Watch the video on Binomial Theorem General Term • Answer the following questions; 1. Write the coefficients of the terms indicated in the expansions of the following a) $(2x + 3)^6$, 6th term b) $(-x + 2)^{20}$, 20th term 2. Use Binomial Theorem to find the value of a) $(1.01)^{10}$, 3 dp b) $(2.001)^{10}$, 6 sigf
Interaction begins	• Post your answers on discussion forum 4.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Binomial Theorem for any index

4.2.2 Binomial Theorem for any Index

Theorem 4.2

For any rational number n , the binomial theorem is given by

$$\begin{aligned}
 (1+x)^n = & 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 \\
 & + \dots,
 \end{aligned}$$

provided $-1 \leq x \leq 1$ i.e. $|x| \leq 1$.

Example 4.6

Expand $(1-x)^{-1}$ in ascending powers of x as far as the term in x^3 .

Solution

$$(1-x)^{-1} = (1-x)^n \quad \text{where } n = -1$$

Therefore

$$\begin{aligned} (1-x)^{-1} &= 1 + (-1)(-x) + \frac{(-1)(-2)}{2!}(-x)^2 + \frac{(-1)(-2)(-3)}{3!}(-x)^3 + \dots \\ &= 1 + x + x^2 + x^3 + \dots, \quad \text{provided } |x| \leq 1. \end{aligned}$$

Example 4.7

Obtain the first five terms of the expansion $\sqrt{1+2x}$ in ascending powers of x . State the values of x for which the expansion is valid.

Solution

$$\sqrt{1+2x} = (1+2x)^{\frac{1}{2}}, \quad n = \frac{1}{2}$$

Therefore

$$\begin{aligned} (1+2x)^{\frac{1}{2}} &= 1 + \frac{1}{2}(2x) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(2x)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(2x)^3 + \\ &\quad \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{4!}(2x)^4 + \dots \\ &= 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 - \frac{5}{8}x^4 + \dots \end{aligned}$$

The expansion is valid for $-1 \leq 2x \leq 1$ ie $|x| \leq \frac{1}{2}$.

Example 4.8

Expand $\frac{1}{(2+x)^2}$ in ascending powers of x as far as the term in x^3 and state the values of x for which the expansion is valid.

Solution

$$\begin{aligned} \frac{1}{(2+x)^2} &= (2+x)^{-2} \\ &= 2^{-2} (1 + \frac{x}{2})^{-2} \\ &= \frac{1}{4} (1 + \frac{x}{2})^{-2} \\ &= \frac{1}{4} \left[1 + (-2) \left(\frac{x}{2} \right) + \frac{(-2)(-3)}{2!} \left(\frac{x}{2} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(\frac{x}{2} \right)^3 + \dots \right] \\ &= \frac{1}{4} \left[1 - x + \frac{3}{2} x^2 - \frac{1}{2} x^3 + \dots \right] \\ &= \frac{1}{4} - \frac{1}{4} x + \frac{3}{8} x^2 - \frac{1}{8} x^3 + \dots \end{aligned}$$

For the expansion to be valid,

$$-1 \leq 2x \leq 1 \text{ ie } |x| \leq \frac{1}{2}.$$

Example 4.9

Expand $\frac{\sqrt{1+2x}}{1-x}$ up to the term including x^3 .

Solution

$$\sqrt{1+2x} = (1+2x)^{1/2} (1-x)^{-1}$$

1-x Now,

$$\begin{aligned} (1+2x)^{1/2} &= 1 + \frac{1}{2}(2x) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(2x)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(2x)^3 + \dots \\ &= 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \dots \end{aligned}$$

and

$$\begin{aligned} (1-x)^{-1} &= 1 + (-1)(-x) + \frac{(-1)(-2)}{2!}(-x)^2 + \frac{(-1)(-2)(-3)}{3!}(-x)^3 + \dots \\ &= 1 + x + x^2 + x^3 + \dots \end{aligned}$$

Therefore

$$\begin{aligned} \sqrt{1+2x} &= (1+2x)^{1/2} (1-x)^{-1} \\ &= \left(1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \dots\right) (1 + x + x^2 + x^3 + \dots) \\ &= \left(1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \dots\right) + \left(x + x^2 + x^3 + \dots\right) + \left(-\frac{1}{2}x^2 - \frac{1}{2}x^3 + \dots\right) + \dots \\ &= 1 + 2x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \dots \end{aligned}$$

Example 4.10

Find the first four terms in the expansion of $(1-8x)^{1/2}$ in descending powers of x . Substitute $x = \frac{1}{100}$ and obtain the value of $\sqrt{23}$ correct to five significant figures.

Solution

$$(1-8x)^{1/2} = 1 + \frac{1}{2}(-8x) + \frac{1}{2}(-1) \frac{(-8x)^2}{2!} + \frac{1}{2}(-1)(-3/2) \frac{(-8x)^3}{3!} + \dots$$

$$= 1 - 4x - 8x^2 - 32x^3$$

Substituting x with $\frac{1}{100}$ we have,

$$(1-8x)^{1/2} = 1 - 4x - 8x^2 - 32x^3$$

$$= 1 - 4\left(\frac{1}{100}\right) - 8\left(\frac{1}{100}\right)^2 - 32\left(\frac{1}{100}\right)^3$$

$$= 1 - 0.04 - 0.0008 - 0.000032$$

$$= 0.959168$$

$$\frac{2}{10}\sqrt{23} = \frac{0.959168 \times 10}{2} = 4.79584$$

$$\sqrt{23} = 4.7958$$

Also

$$(1-8x)^{1/2} = 1 - 4x - 8x^2 - 32x^3$$

$$= 1 - 4(0.01) - 8(0.01)^2 - 32(0.01)^3$$

$$= 1 - 0.04 - 0.0008 - 0.000032$$

$$= 0.959168$$

$$\frac{2}{10}\sqrt{23} = \frac{0.959168 \times 10}{2} = 4.79584$$

$$\sqrt{23} = 4.7958$$

Etivity 4.2.2: Binomial Theorem for any index

Numbering and pacing and sequencing	4.2.2
Title	Binomial Theorem for any index
Purpose	To enable you to carry out binomial expansion for any index.
Brief summary of overall task	Watch the video on Binomial Theorem for any index by Doubtnut and answer the given questions.
Spark	<i>Binomial theorem:</i> $(1+x)^n = 1 + nx + \frac{n(n-1)}{2} x^2 + \dots$
Individual contribution	- Watch the video on Binomial Theorem for any index - Answer the following questions; Expand the following in ascending powers of x as far as the term in x^3 a) $(1+x)^{1/3}$ b) $\frac{1}{\sqrt{2+x^2}}$ c) $(1+x)^{-1}$
Interaction begins	- Post your answers on discussion forum 4.2.2 - Read what your colleagues have posted. - In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	- Focussing group discussion - Encouraging lurkers (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This activity should take one hour.
Next	Introduction to set theory

4.3 Assessment

1. Expand $(1 + 2x - x^2)$ as far as the terms in x^3 .

2. Obtain the first four terms of the expansion of $(1 + 8x)^2$ in ascending powers of x .

Putting $x = \frac{1}{100}$ obtain the value of $\sqrt[3]{100}$ correct to five decimal places.

3. Show that if x is small enough for its cube and higher powers to be neglected,

$$\sqrt{1-x} \approx 1 - x + x^2.$$

4.4 References

1. Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
2. Bostock, L., Chandler, S. and Rourke, C. (1982). Further Pure Mathematics. Stanley Thornes Publishers Ltd.
3. Larson, R. and Hostetler, R. (1997). Precalculus, Houghton Company.

LECTURE FIVE

SET THEORY

5.1 Introduction

In this lesson we introduce sets theory, Venn diagrams and applications to counting. The concept of set is fundamental to mathematics and computer science. Everything mathematical starts with sets. For example, relationships between two objects are represented as a set of ordered pairs of objects, the natural numbers, which are the basis of other numbers, are also defined using sets, graphs and digraphs consisting of lines and points are described as an ordered pair of sets and so on.

5.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) Discuss basic concepts in set theory
- ii) Perform set operations
- iii) Simplify set

expressions using set laws iv) Apply set theory to counting.

5.2.1 Introduction to Sets

Definition 5.1

A set is a well-defined list or collection of objects. The different objects that form a set are called members or elements of the set. The members making up a set are enclosed in braces (curly brackets) i.e. $\{ \}$. We usually denote sets by capital letters. For example $A = \{1, 10, 5, 3\}$ is a set having four members.

A set is finite or infinite according to whether it has a finite or infinite number of members. For example

- i) $A = \{a, b, c, d\}$ is a finite set. ii)
 $N = \{1, 2, 3, \dots\}$ is an infinite set.

There are two ways of describing a set;

- i) By listing down all of its members. ii) By stating the properties characterizing each one of the members.

For example

- a) $A = \{a, b, c, d, e, f\}$ denotes a set A whose members are a, b, c, d, e and f .
 b) $B = \{x : x \geq 0 \text{ and } x \text{ is an integer}\}$. Then the members of B are $1, 2, 3, \dots$
 c) $E = \{x \mid x^2 - 3x + 2 = 0 \text{ and } x \text{ is a real number}\}$. Then the members of E are all real numbers satisfying the equation $x^2 - 3x + 2 = 0$, that is $E = \{1, 2\}$.

Null or Empty Set

It is a set with no members and it is denoted by $\emptyset$ or $\{ \}$.

Example 5.1

- a) The set $\{x \mid x^2 + 1 = 0 \text{ and } x \text{ is a real number}\}$ is empty for $x^2 + 1 = 0$ has no real solution.
 b) The set $\{x : x \text{ is an even integer lying between } 10 \text{ and } 11, 10 \text{ not included}\}$ is empty.

Membership of a Set

The symbol $\in$ means is a 'member of' or 'belong to'.

Thus if $A = \{1, 5, 6, 7\}$, then $1 \in A$.

The symbol $\notin$ means 'is not a member of' or 'doesn't belong to'. Thus if $B = \{a, b, e, d\}$, then $x \notin B$.

Equality of Sets

Two sets A and B are said to be equal, written $A = B$, if they have exactly the same members.

Remark 5.1

The order in which members of a set are listed down is immaterial. For example

$A = \{a, b, c, d, e, f\}$ and $B = \{f, d, e, a, c, b\}$ are equal.

Remark 5.2

It is not allowed to repeat a member of a set more than once.

Thus if $A = \{a, b, a, d, e, e, f\}$ and $B = \{a, b, d, e, f\}$, then $A = B$.

Cardinality of a set

The cardinality or the order of a set A is the number of elements in A . It is denoted by $n(A)$ or $|A|$. For example if $B = \{3, 7, 8, 9, 11\}$, then $n(B) = 5$.

Subsets

If every member of a set A is also a member of a set B , then A is said to be a subset of B or A is contained in B . If A is contained in B , we denote this by $A \subseteq B$. If A is a subset of B , then B is called a superset of A . The notation $A \subset B$ means A is contained in B but $A \neq B$. In this case, A is called a proper subset of B . Furthermore the symbol $A \not\subseteq B$ means A is not contained in B ,

thus we can find $x \in A$ but $x \notin B$.

Example 5.2

Find all the subsets of $A = \{1, 2, 3\}$.

Solution

a) $\emptyset$ b) $\{1\}$ c) $\{2\}$ d) $\{3\}$ e) $\{1, 2\}$ f) $\{1, 3\}$ g) $\{2, 3\}$ h) $A = \{1, 2, 3\}$

Remark 5.3

The empty set $\emptyset$ is a subset of any other set since it has no members. The subsets a) to g) are proper subsets of A .

If A contains an element which is not in B , the A cannot be a subset of B . We denote this by $A \not\subset B$.

Example 5.3

If $A = \{1, 2, 4, 5\}$ and $B = \{4, 5, 7, 8\}$, then $A \not\subset B$ since $1, 2 \in A$ but $1, 2 \notin B$.

If $n(A) = m$, then the number N of all subsets of A is $N = 2^m$.

Example 5.4

If $n(A) = 3$, then the number of all subsets of A is $2^3 = 8$.

Equality of Sets

A set A is said to be equal to B written $A = B$ if $A \subset B$ and $B \subset A$. Therefore in order to prove that $A = B$ we need to show that $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$.

Universal Set

If we have some sets under consideration, a fixed set which contains all these subsets is called the universal set and it is denoted by U .

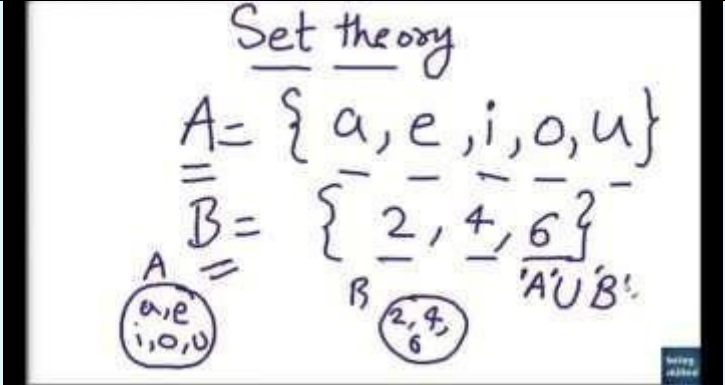
Example 5.5

Let $A = \{1, 2, 3, 4\}$ and $B = \{5, 7, 9\}$ and $C = \{10, 12, 13\}$. Then we can take the universal set to be either

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$, $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$, $N = \{1, 2, 3, \dots\}$

or $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

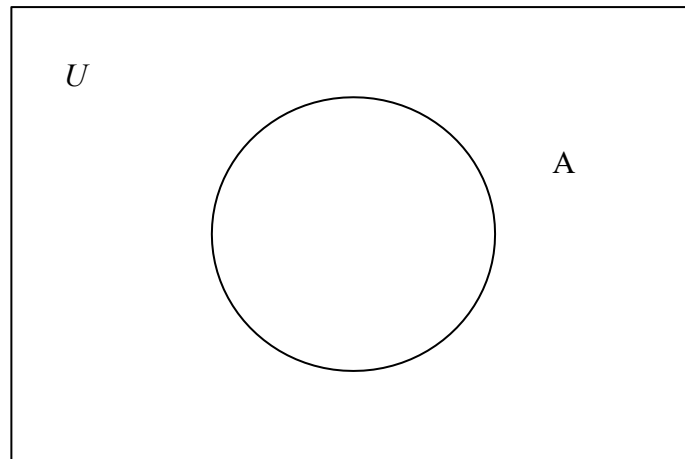
E-tivity 5.2.1: Introduction to sets

Numbering and pacing and sequencing	5.2.1
Title	Introduction to sets
Purpose	To expose you to various basic concepts in set theory so that you can be able to perform set operations.
Brief summary of overall task	Watch the video on introduction to set theory Jeff Suzuki and then Discuss the basic concepts in set theory
Spark	
Individual contribution	• Watch the video on introduction to set theory • Reflect on the ways set theory can be applied in real life • Explain the basic concepts in set theory
Interaction begins	• Post your answers on the discussion forum 5.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Summarising key points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	Venn diagrams and set operations

5.2.2 Venn Diagrams and Set Operations

5.2.2.1 Venn Diagrams

Set and set problems may be represented by diagrams called Venn diagrams. The universal set is represented by a rectangle while subsets of the universal set are represented by circles or discs lying inside the rectangle as illustrated below;

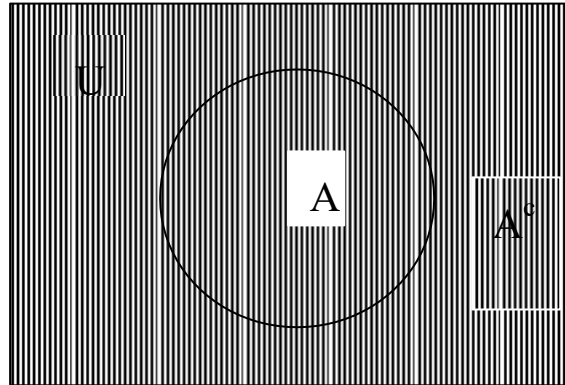


Complement of a set

If U is the universal set and A is a subset of U , then all elements of U which do not belong to A form the complement of A . The complement of A is denoted by A^c . Thus

$$A^c = \{x : x \in U \text{ and } x \notin A\}.$$

Diagrammatically,



Shaded area represents A^c .

Example 5.6

Let $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 3, 5\}$. Then

$$A^c = \{2, 4, 6\}.$$

5.2.2.2 Set Operations

We consider the following set operations;

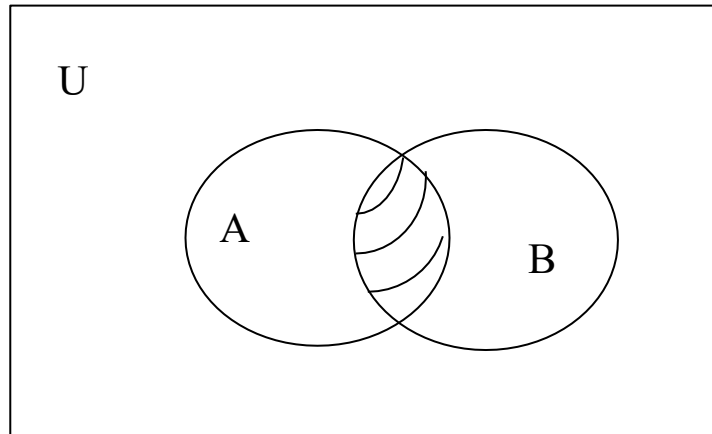
Intersection

Let A and B be two sets. Then the set of elements which are common to both A and B is called the intersection of A and B , denoted by $A \cap B$.

Thus

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Diagrammatically,



The shaded region represents $A \cap B$.

Examples 5.7

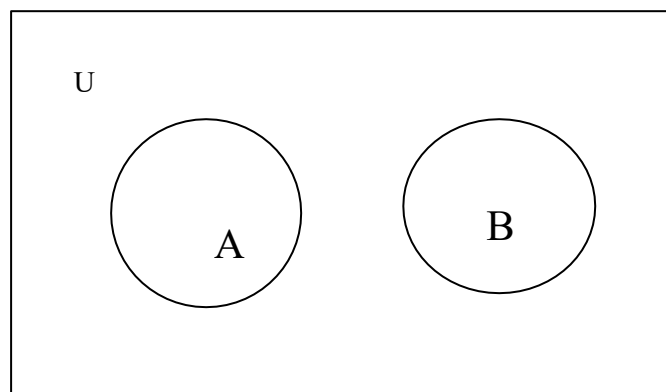
If $A = \{1, 3, 5, 7, 9\}$ and $B = \{7, 9, 11\}$, find $A \cap B$.

Solution

$$A \cap B = \{7, 9\}$$

If A and B have no elements in common i.e. $A \cap B = \emptyset$, then A and B are said to be disjoint.

Diagrammatically,



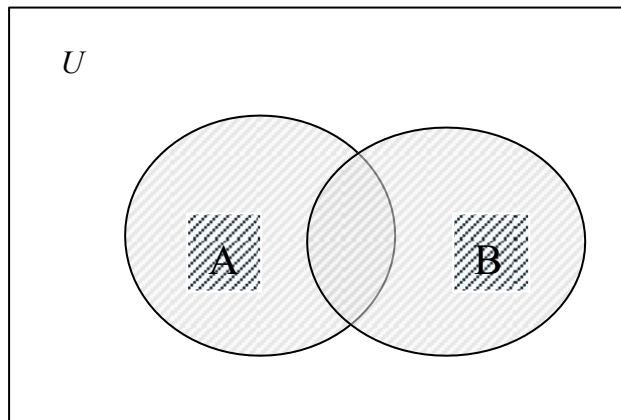
Example 5.8

If $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8\}$, then A and B are disjoint since $A \cap B = \emptyset$.

Union

The union of two sets A and B is the set of all elements contained in A or B . The union of two sets A and B is denoted by $A \cup B$ (read A union B). Therefore $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

Diagrammatically,



Shade area represents $A \cup B$.

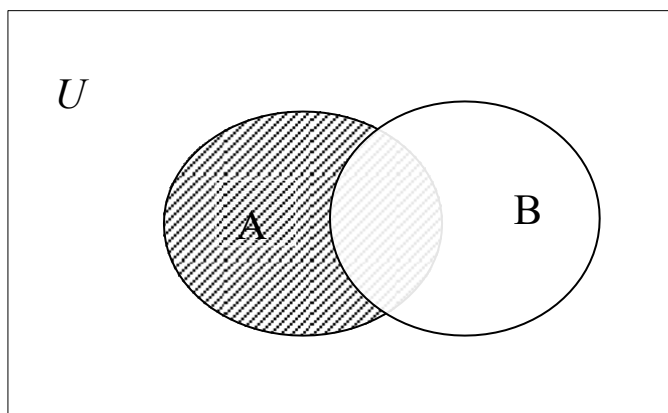
Difference of Two Sets

The relative complement of a set B with respect to a set A , or simply the difference of A and B denoted by $A \setminus B$ (or $A - B$) is the set of elements which belong to A but do not belong to B .

Therefore

$$A \setminus B = A - B = \{x \mid x \in A, x \notin B\}.$$

Diagrammatically,

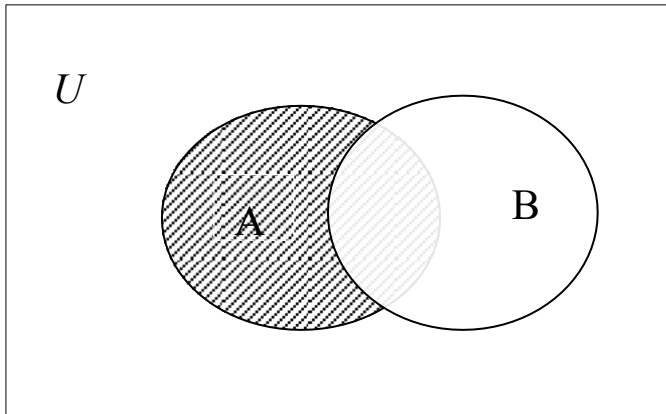


Shaded area is $A \setminus B$.

Example 5.9

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 6\}$. Then

a) $A \setminus B = \{1, 2\}$ and b) $B \setminus A = \{6\}$



Shaded area is $A \setminus B$.

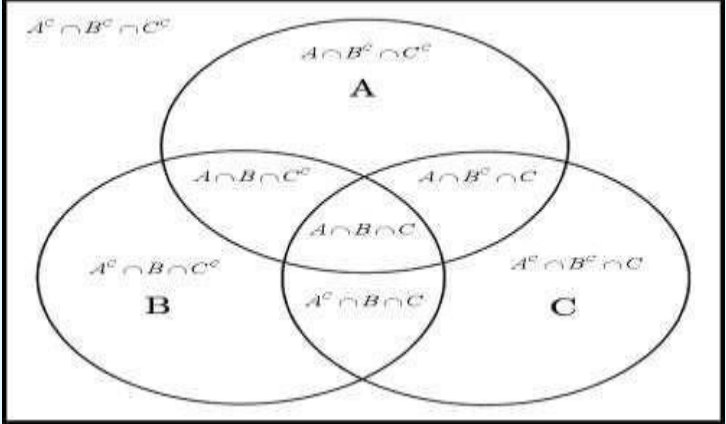
Example 5.9

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 6\}$. Then

a) $A \setminus B = \{1, 2\}$ and b) $B \setminus A = \{6\}$

E-tivity 5.2.2: Venn diagrams and set operations

Numbering and pacing and sequencing	5.2.2
Title	Venn diagrams and set operations

Purpose	To expose you to various set operations and pictorial representation of sets.
Brief summary of overall task	Watch the video on Set and set operations by GVSU Math and solve problems involving set operations.
Spark	Venn diagram involving three sets 
Individual contribution	• Watch the video on set and set operations • Answer the question; Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$ Find (i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$ (iv) $B - A$
Interaction begins	• Post your answers on the discussion forum 5.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Summarising key points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	Algebra of sets

5.2.3 Algebra of sets

Theorem 5.1 (Set Laws)

Let A , B and C be sets. Then

1. $A \cap A = A$ $A \cup A = A$ Idempotent Law
2. $(A \cap B) \cap C = A \cap (B \cap C)$ $(A \cup B) \cup C = A \cup (B \cup C)$ Associative Law
3. $A \cap B = B \cap A$ $A \cup B = B \cup A$ Commutative Law
4. a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ Distributive Law
5. a) $A \cap \emptyset = \emptyset$
 b) $A \cap U = A$
 c) $A \cup \emptyset = A$ Identity Laws
 d) $A \cup U = U$
6. $(A^c)^c = A$ Involution Law
7. a) $A \cap A^c = \emptyset$
 b) $A \cup A^c = U$ Complement Laws
 c) $\emptyset^c = U$
 d) $U^c = \emptyset$
8. a) $(A \cap B)^c = A^c \cup B^c$ De Morgan's Laws b) $(A \cup B)^c = A^c \cap B^c$

Remark 5.4

To prove equations involving set operations we use definition of equality of two sets. The alternative method of proof involves use of a Venn diagram.

Example 5.10

We prove the first of the De Morgan's Laws.

Proof**Method 1**

We first show that $(A \cap B)^c \subseteq A^c \cap B^c$. If $x \in (A \cap B)^c$, then $x \notin (A \cap B)$. Thus $x \notin A$ and $x \notin B$, so that $x \in A^c$ and $x \in B^c$. Hence $x \in A^c \cap B^c$.

Next we show that $A^c \cap B^c \subseteq (A \cap B)^c$. Let $x \in A^c \cap B^c$, then $x \in A^c$ and $x \in B^c$, so $x \notin A$ and

$x \notin B$. Hence $x \notin (A \cap B)$, so $x \in (A \cap B)^c$. Together, these inclusions prove that $(A \cap B)^c = A^c \cap B^c$. $\square$

Method 2

Using Venn Diagrams

Figure I**Figure II**

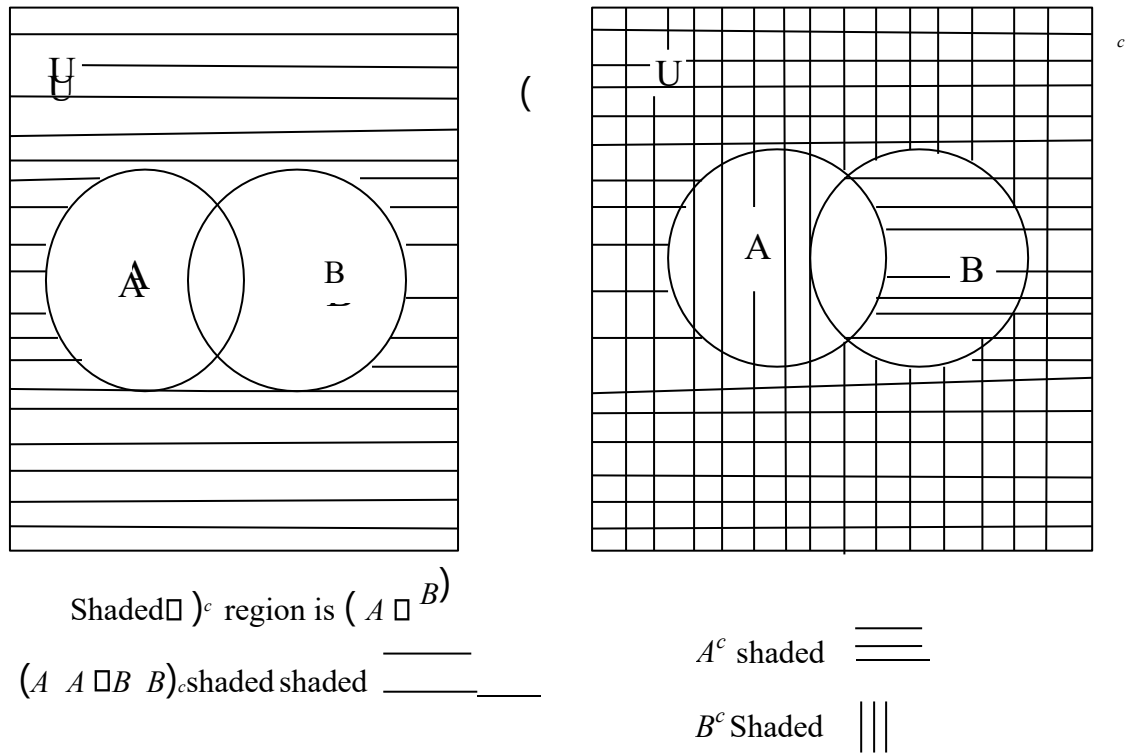
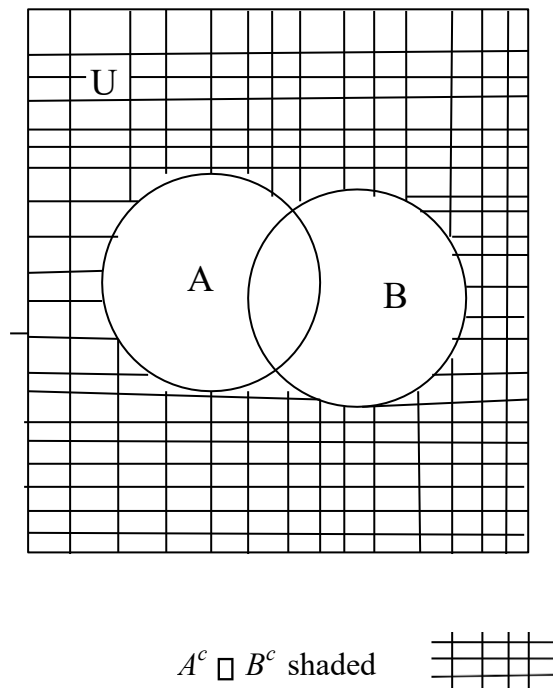


Figure III



Since the shaded region in Figures I and III are identical then, $(A \cup B)^c = A^c \cap B^c$.

Example 5.11

Use set algebra to simplify

a) $A \cap (A^c \cap B)$

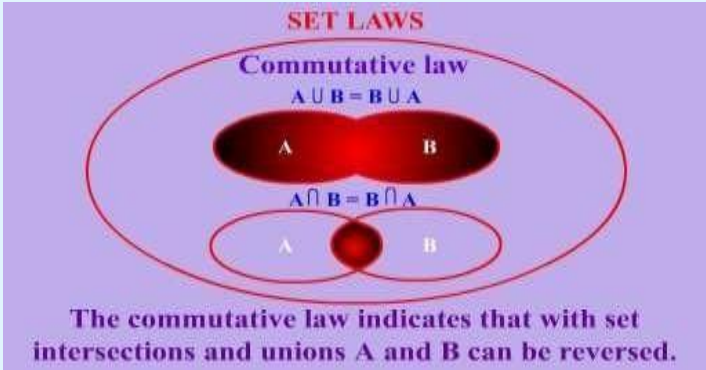
b) $(A \cap B) \cap (A \cap B^c)$

Solution

a) $A \cap (A^c \cap B) = (A \cap A^c) \cap (A \cap B) = U \cap (A \cap B)$
 $= A \cap B.$

b) $(A \cap B) \cap (A \cap B^c) = A \cap (B \cap B^c)$
 $= A \cap U$
 $= A.$

E-tivity 5.2.3: Algebra of sets

Numbering and Pacing and Sequencing	5.2.3
Title	Algebra of sets
Purpose	To expose you to algebra of sets so that you can be able to simplify set expressions.
Brief summary of overall task	Watch the video Laws of set theory by Mouna Naravani and then simplify set expressions.
Spark	
Individual contribution	• Watch the video on Laws of set theory • - Simplify the following set operations; a) $A \cap (A^c \cap B)$ b) $(S \cap T) \cap (S^c \cap T) \cap (S \cap T^c)$

Interaction begins	• Post your answers on the discussion forum 5.2.3 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Summarising key points • Closing the discussion
Schedule and time	This activity should take one hour
Next	Applications of set theory to counting

5.2.4 Applications of Set Theory to Counting

A set is finite or infinite according to whether it has a finite or infinite number of elements.

For example

- a) $A = \{a, b, c\}$ is finite since $n(A) = 3$.
- b) $N = \{1, 2, 3, \dots\}$ is an infinite set.

Lemma 5.2

If A and B are disjoint finite sets, then $A \cup B$ is finite and n

$$(A \cup B) = n(A) + n(B).$$

Proof

In counting elements of $A \cup B$, first count those that are in A . There are $n(A)$ of these. The only other elements of $A \cup B$ are those that are in B but not in A . But since A and B are disjoint, no elements of B is in A , so there are $n(B)$ elements that are in B and not in A . Therefore $n(A \cup B) = n(A) + n(B)$.

Theorem 5.3 [Inclusion-Exclusion Principle]

Let A and B be any finite sets, then $A \cup B$ is finite and n

$$(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Proof

This follows from the fact that when we add $n(A)$ and $n(B)$, we have counted the elements of $A \cap B$ twice.

This Principle holds for any number of sets.

For three sets we have

Theorem 5.4

For any finite sets A, B, C we have $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

Example 5.12

Among 50 students in a class, 26 got an A in the first examination and 21 got an A in the second examination. If 17 students did not get an A in either examination, how many students got A in both the examinations?

Solution

Let P be the set of students who scored A in the first exam, then $|P| = 26$

Let Q be the set of students who scored A in the second exam, then $|Q| = 21$ Also

$$|U| = 50 \text{ and } |(P \cup Q)^c| = 17. \text{ Therefore } |P \cup Q| = 50 - 17 = 33.$$

But

$$|P \cup Q| = |P| + |Q| - |P \cap Q|$$

and so

$$33 = 26 + 21 - |P \cap Q|$$

$$\therefore |P \cap Q| = 26 + 21 - 33 = 14$$

Hence 14 Students scored an A in both exams.

Example 5.13

In a survey of 60 people, it was found that

25 people read the Newsweek magazine

26 read Time

26 read Fortune

9 read both Newsweek and Fortune

11 read both Newsweek and Time

8 read both Time and Fortune

3 read all the three magazine

- Find the number of people who read at least one of the three magazines.
- Fill in the correct number of people in a Venn diagram.
- Find the number of people who read exactly one magazine.

Solution

- We need to find $n(N \cup T \cup F)$. Now $n(N \cup T \cup F) = n(N) + n(T) + n(F) - n(N \cap T) - n(N \cap F) - n(T \cap F) + n(N \cap T \cap F)$

$$= 25 + 26 + 26 - 11 - 9 - 8 + 3 = 52$$

- The required Venn diagram is obtained as follows;

3 read all the three magazines

$11 - 3 = 8$ read Newsweek and Time but not all the three magazines

$9 - 3 = 6$ read Newsweek and Fortune but not all the three magazines

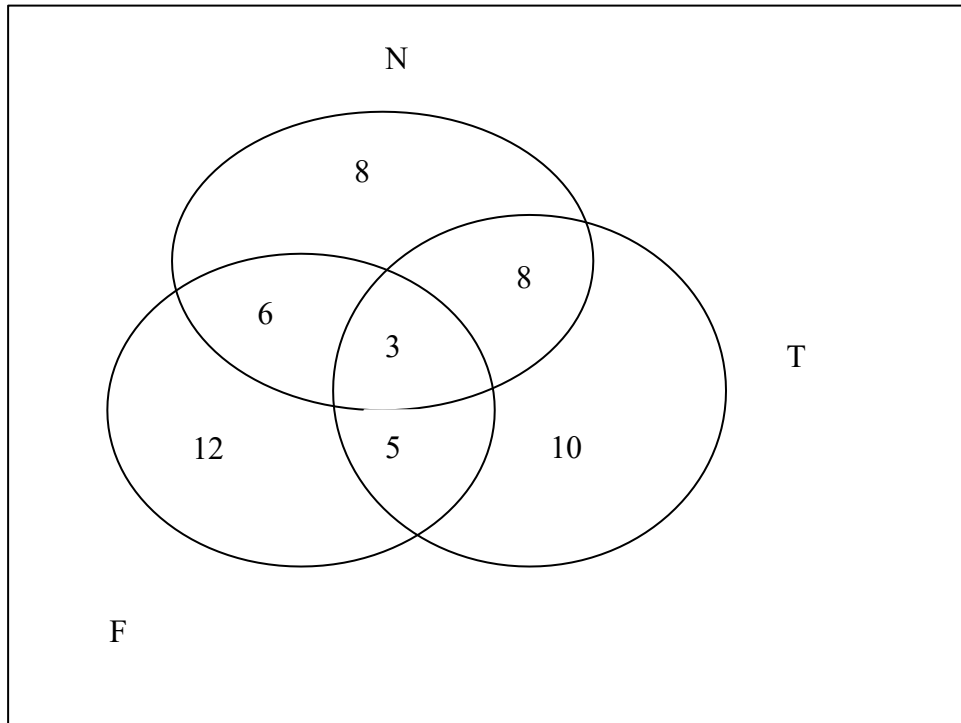
$8 - 3 = 5$ read Time and Fortune but not all the three magazines

$25 - 8 - 6 - 3 = 8$ read only Newsweek

$26 - 8 - 5 - 3 = 10$ read only Time

$26 - 6 - 5 - 3 = 12$ read only Fortune

$60 - 52 = 8$ read no magazine at all



c) $8+10+12 = 30$ read only one magazine.

E-tivity 5.2.4: Applications of set theory to counting

Numbering, pacing and sequencing	5.2.4
Title	Applications of set theory to counting
Purpose	To expose you to applications of set theory to counting so that you can be able to solve real life problems.
Brief summary of overall task	Watch videos on Applications of set theory Part 1 and Applications of set theory Part 3 by Integrity Gurukul and then solve counting problems.
Spark	Applications of set theory

Individual contribution	• Watch videos on applications of set theory part 1 and 3 • Answer the question; There are 30 students in a class. Among them, 8 students are learning both English and French. A total of 18 students are learning English. If every student is learning at least one language, how many students are learning French in total?
Interaction begins	• Post your answers on the discussion forum 5.2.4 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Summarising key points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	Logic

5.3 Assessment

1. In a class of 80 students, 50 students know English, 55 know French and 46 know German language. 37 students know English and French, 28 students know French and Germany, 25 students know English and Germany and 7 students know none of the languages. Find out
 - a) How many students know all the three languages?
 - b) How many students know exactly 2 languages?
 - c) How many know only one language?
2. Prove the following identities

i) $(A \cap B^c) \cap (A^c \cap B) \cap (A \cap B) = A \cap B.$

ii) $A - (B \cap C) = (A - B) \cup (A - C)$

5.4 References

1. Goldstein, L., Schneider, D. and Siegel, M., Finite Mathematics and its applications, 7th Ed., Prentice Hall, 1998.

2. Grimaldi, R.P., Discrete and Combinatorial Mathematics. An applied introduction, 5th Ed., Pearson Addison Wesley, 2004.
3. Edgar, G., Goodaire, M., Parmenter, M., Discrete Mathematics with graph theory, 2nd Ed., 2002.
4. Seymour Lipschutz and Marc Lipson. Schaum's Outline Series Discrete Mathematics, 3rd edition, McGraw-Hill Companies, Inc., 2007.
<https://alas.matf.bg.ac.rs/~mi10164/Materijali/DS.pdf>

LECTURE SIX

LOGIC

6.1 Introduction

In this lesson we will study Logic and logical operations. Logic is a language for reasoning. It is a collection of rules we use when doing logical reasoning. Human reasoning has been observed over centuries from at least the times of Greeks, and patterns appearing in reasoning have been extracted, abstracted, and streamlined. In logic we are interested in true or false of statements, and how the truth/falsehood of a statement can be determined from other statements. However, instead of dealing with individual specific statements, we are going to use symbols to represent arbitrary statements so that the results can be used in many similar but different cases. The formalization also promotes the clarity of thought and eliminates mistakes.

6.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) Explain what is meant by a proposition.
- ii) Perform logical operations on propositions
- iii) Construct truth tables for propositions
- iv) Determine truth values of conditional and biconditional statements.

6.2.1 Introduction to Propositions

Many proofs in Mathematics and many algorithms in Computer Science use logical expressions of the form

'IF p THEN q ' or 'IF p_1 AND p_2 , THEN q_1 OR q_2 '.

It is necessary to know the cases in which these expressions are either TRUE or FALSE; what we refer to as the truth values of such expression.

Definition 6.1

A proposition (or a statement) is a declarative sentence which is either true or false but not both. Consider the following eight sentences;

a) Paris is in France

b) $1+1 = 2$

- c) $2+2=3$
- d) London is in Denmark
- e) $9 < 6$
- f) $x = 2$ is a solution of $x^2 = 4$
- g) Where are you going?
- h) Do your homework

All are propositions except g) and h). Moreover a), b) and f) are true whereas c), d) and e) are false.

6.2.1.1 Compound propositions

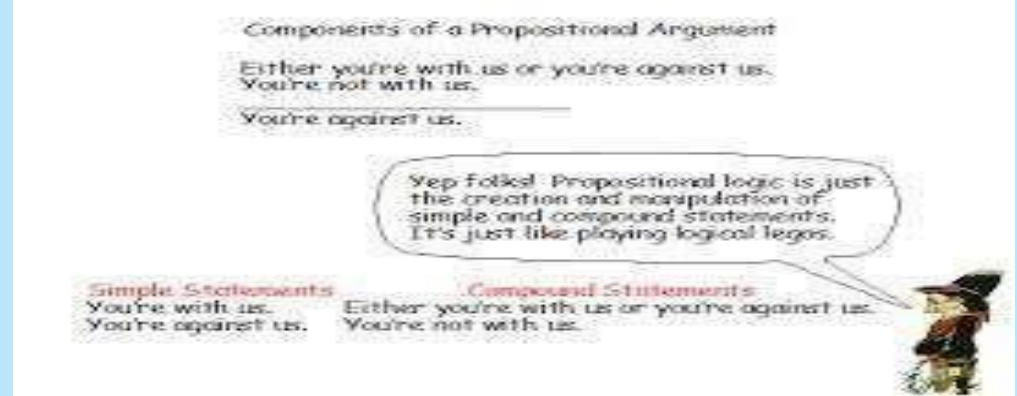
A logical connective is a symbol or word used to connect two or more propositions in a grammatically valid way, such that the sense of the resulting proposition (called compound proposition) depends only on the original proposition. A proposition which cannot be broken down into simpler propositions is called a primitive proposition.

Example 6.1

- a) Roses are red and violets are blue is a compound proposition with sub propositions ‘Roses are red’ and ‘Violets are blue.’
- b) John is intelligent or studies every night is a compound proposition with sub propositions ‘John is intelligent’ and ‘John studies every night.’

E-tivity 6.2.1: Introduction to Propositions

Numbering and pacing and sequencing	6.2.1
Title	Introduction to Propositions
Purpose	To introduce you to propositional logic
Brief summary of overall task	Watch the video on Introduction to propositional logic by Digiimento and answer the given questions.

Spark	
Individual contribution	• Watch the video on introduction to propositional logic • Explain in your words giving examples the concepts of proposition and compound proposition
Interaction begins	• Post your answers on discussion forum 6.2.1 • Read what your colleagues have posted.
	• In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour.
Next	Basic Logical Operations

6.2.2 Basic Logical Operations

Conjunction $p \sqcap q$

Any two propositions p and q can be combined by the word ‘and’ denoted $\sqcap$ to form a compound proposition called conjunction of the original propositions. Symbolically this is written

$$p \sqcap q,$$

and is read ‘ p and q ’. Since $p \sqcap q$ is a proposition, it has truth values that depend only on truth values of p and q .

Definition 6.3

If p and q are true, then $p \sqcap q$ is true; otherwise $p \sqcap q$ is false.

The truth values of $p \sqcap q$ may be defined equivalently by the following table:

$$p \sqcap q$$

p	q	$p \sqcap q$
T	T	T
T	F	F
F	T	F
F	F	F

The four lines correspond to the four possible combinations of T and F for sub propositions p and q .

Example 6.2

Consider the following statements

i) Paris is in France and $2+2 = 4$ ii)

Paris is in France and $2+2 = 5$ iii)

Paris is in England and $2+2 = 4$ iv)

Paris is in England and $2+2 = 5$

Only the first statement is true. Each of the other statements is false since at least one of its sub statements is false.

Disjunction $p \sqcup q$

Any two propositions p and q can be combined by the word ‘or’ denoted $\sqcup$ to form a compound proposition called disjunction of the original propositions. Symbolically this is written

$$p \sqcup q$$

and is read as p or q . The truth values of $p \sqcup q$ depends only on the truth values of p and q as follows;

Definition 6.4

If p and q are false, then $p \sqcup q$ is false; otherwise $p \sqcup q$ is true. Truth value of $p \sqcup q$ may be defined equivalently by the table;

$$p \sqcup q$$

p	q	$p \sqcup q$
-----	-----	--------------

T	T	T
T	F	T
F	T	T
F	F	F

Example 6.3

Consider the following statements

i) Paris is in France or $2+2 = 4$ ii)

Paris is in France or $2+2 = 5$ iii)

Paris is in England or $2+2 = 4$ iv)

Paris is in England or $2+2 = 5$

Only the last statement is false. Each of the other statements is true since at least one of its sub statements is true.

Negation $\sim p$

Given any proposition p , another proposition called negation of p , can be formed by writing ‘it is not the case that’ or ‘it is false that ...’ before p or if possible by inserting in p the word not.

Symbolically this is written

$$\sim p$$

and is read ‘not p ’ or negation of p .

Definition 6.5

If p is true, then $\sim p$ is false, and if p is false, then $\sim p$ is true.

$$\sim p$$

p	$\sim p$
T	F
F	T

E-tivity 6.2.2: Basic Logical Operations

Numbering and pacing and sequencing	6.2.2																		
Title	Basic Logical Operations																		
Purpose	To introduce you to connectives and basic logical operations																		
Brief summary of overall task	Watch the video on Propositional logic: operations by Math Forge and answer the given questions.																		
Spark	<table border="1"><thead><tr><th></th><th><i>Symbol</i></th><th><i>Meaning</i></th></tr></thead><tbody><tr><td>Negation</td><td>$\sim$</td><td>NOT</td></tr><tr><td>Conjunction</td><td>$\wedge$</td><td>AND</td></tr><tr><td>Disjunction</td><td>$\vee$</td><td>OR</td></tr><tr><td>Conditional</td><td>$\rightarrow$</td><td>If/Then</td></tr><tr><td>Biconditional</td><td>$\longleftrightarrow$</td><td>IF and ONLY IF</td></tr></tbody></table>		<i>Symbol</i>	<i>Meaning</i>	Negation	$\sim$	NOT	Conjunction	$\wedge$	AND	Disjunction	$\vee$	OR	Conditional	$\rightarrow$	If/Then	Biconditional	$\longleftrightarrow$	IF and ONLY IF
	<i>Symbol</i>	<i>Meaning</i>																	
Negation	$\sim$	NOT																	
Conjunction	$\wedge$	AND																	
Disjunction	$\vee$	OR																	
Conditional	$\rightarrow$	If/Then																	
Biconditional	$\longleftrightarrow$	IF and ONLY IF																	
Individual contribution	• Watch the video on Propositional logic: operations • Determine the truth values of each of the following statements; a) $4+2 = 5$ and $6+3 = 9$ b) $3+2 = 5$ or $6+1 = 7$																		
Interaction begins	• Post your answers on discussion forum 6.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind																		
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion																		
Schedule and time	This activity should take one hour.																		
Next	Propositions and truth tables																		

6.2.3 Propositions and Truth Tables

Let $P(p_1, p_2, \dots, p_n)$ denote an expression constructed from the logical variables $p_1, p_2, \dots, p_n$, which take truth value TRUE (T) or FALSE (F) and various logical connectives $\neg$, $\wedge$ and $\sim$. Such an expression $P(p_1, p_2, \dots, p_n)$ will be called a compound proposition. The main property of the proposition $P(p_1, p_2, \dots, p_n)$ is that its truth value depends exclusively upon the truth values of the variable. A simple and concise way to show this property is through a truth table.

6.2.3.1 Steps for constructing Truth Table

1. In the first row of the truth table the first n columns are for the variables $p_1, p_2, \dots, p_n$.
Next, ensure that there are enough rows in the table to allow for all possible combinations of T and F for these variables. (For two variables, 4 rows are necessary; for three variables, 8 rows are necessary and in general for n variables, 2^n rows are necessary.)
2. There is then a column for each elementary stage of construction of the proposition, the truth value of each step being determined from previous stages by definition of connectives $\neg$, $\wedge$ and $\sim$.
3. Finally we obtain the truth values of the proposition $P(p_1, p_2, \dots, p_n)$ which appears in the last column.

Example 6.4

Construct the truth table for the proposition $\sim(p \wedge \sim q)$.

Solution

$$\sim(p \wedge \sim q)$$

p	q	$\sim q$	$p \wedge \sim q$	$\sim(p \wedge \sim q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

Remark 6.1

We adopt the following order of precedence for logical connectives;

$\sim$ has precedence over $\square$ which has precedence over $\square$ **Eg**

$\sim p \square q$ means $(\sim p) \square q$ and not $\sim (p \square q)$.

Example 6.5

Construct the truth table for the proposition $p \square q \square r$.

Solution

By Remark 2.1, the given proposition can be written as $p \square (q \square r)$.

$$p \square (q \square r)$$

p	q	r	$q \square r$	$p \square (q \square r)$
T	T	T	T	T
T	T	F	F	T
T	F	T	F	T
T	F	F	F	T
F	T	T	T	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

6.2.3.2 Tautologies and Contradictions

Some proposition $P(p_1, p_2, \dots, p_n)$ contain only T in the last column of their truth tables or, in other words, they are true for any truth values of their variables. Such propositions are called tautologies.

Analogously, a proposition $P(p_1, p_2, \dots, p_n)$ is called a contradiction if it contains only F in the last column of its truth table, in other words, it is false for any truth values of its variables.

Example 6.6

The proposition p or not p , that is $p \sqcup \sim p$ is a tautology, and the proposition p and not p , that is $p \sqcap \sim p$ is a contradiction.

a) $p \sqcup \sim p$

b) $p \sqcap \sim p$

p	$\sim p$	$p \sqcup \sim p$
T	F	T
F	T	T

p	$\sim p$	$p \sqcap \sim p$
T	F	F
F	T	F

Example 6.7

Determine whether or not $p \sqcup q \sqcap p$ is a tautology or contradiction.

Solution

The given proposition can be written as $p \sqcup (q \sqcap p)$

$$p \sqcup (q \sqcap p)$$

p	q	$q \sqcap p$	$p \sqcup (q \sqcap p)$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

Hence $p \sqcup (q \sqcap p)$ is neither a tautology nor a contradiction.

6.2.3.3 Logical Equivalence

Two propositions $P(p_1, p_2, \dots, p_n)$ and $Q(q_1, q_2, \dots, q_n)$ are said to be logically equivalent or equivalent, denoted $P(p_1, p_2, \dots, p_n) \equiv Q(q_1, q_2, \dots, q_n)$ if they have identical truth tables.

Example 6.8

Consider the truth tables of $\sim(p \sqcup q)$ and $\sim p \sqcap \sim q$

a) $\sim(p \sqcup q)$

b) $\sim p \sqcap \sim q$

p	q	$p \sqcup q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$\sim p$	$\sim q$	$\sim p \sqcap \sim q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Observe that the last columns of both tables are the same; that is both propositions are false in the first case and true in the other cases. Accordingly, we write

$$\sim(p \sqcup q) \equiv \sim p \sqcap \sim q.$$

Example 6.9

Show that $\sim(p \sqcup q) \equiv \sim p \sqcap \sim q$.

Solution

Consider the truth tables of $\sim(p \sqcup q)$ and $\sim p \sqcap \sim q$

a) $\sim(p \sqcup q)$

b) $\sim p \sqcap \sim q$

p	q	$p \sqcup q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$\sim p$	$\sim q$	$\sim p \sqcup \sim q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Hence $\sim (p \sqcup q) \sqcup \sim p \sqcup \sim q$.

E-tivity 6.2.3: Propositions and truth tables

Numbering and pacing and sequencing	6.2.3															
Title	Propositions and truth tables															
Purpose	To you to develop skills of constructing truth tables															
Brief summary of overall task Spark	Watch the video on Truth table logic basics by KarByte CS and answer the given questions. <table><tr><th>p</th><th>q</th><th>$p \wedge q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>F</td></tr><tr><td>F</td><td>F</td><td>F</td></tr></table> p and q are true, $p \wedge q$ is true. p or q are true, $p \wedge q$ is false. p and q are false, $p \wedge q$ is false.	p	q	$p \wedge q$	T	T	T	T	F	F	F	T	F	F	F	F
p	q	$p \wedge q$														
T	T	T														
T	F	F														
F	T	F														
F	F	F														
Individual contribution	• Watch the video on Truth table logic basics • Verify that the proposition $p \sqcup \sim (p \sqcup q)$ is a tautology															
Interaction begins	• Post your answers on discussion forum 6.2.3 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind															

E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take one hour
Next	Conditional and biconditional statements

6.2.4 Conditional and Biconditional Statements

Many statements, particularly in Mathematics are of the form ‘if p then q ’. Such statements are called conditional statements and are denoted by $p \rightarrow q$.

The conditional statement $p \rightarrow q$ is frequently read as ‘ p implies q ’.

Another common statement is of the form ‘ p if and only if q ’. Such statements are called biconditional statements and are denoted by $p \leftrightarrow q$.

Definition 6.6

- a) The conditional statement $p \rightarrow q$ is false only if the first part p is true and the second part q is false.

Accordingly, when p is false, the conditional $p \rightarrow q$ is true regardless of the truth value of q .

- b) The biconditional $p \leftrightarrow q$ is true whenever p and q have the same truth values and false otherwise.

a) $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

b) $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Now consider the following truth table of $\sim p \leftrightarrow q$.

c) $\sim p \sqcap q$

p	q	$\sim p$	$\sim p \sqcap q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

Then the truth tables of $\sim p \sqcap q$ and $p \rightarrow q$ are identical, that is, they are both false only in the second case. Accordingly $p \rightarrow q$ is logically equivalent to $\sim p \sqcap q$; that is $p \rightarrow q \sqcap \sim p \sqcap q$.

Example 6.10

Rewrite the following statements without using conditionals.

If it is cold, he wears a hat.

a)

If productivity increases, then the wages rise.

b)

Solution

a) Let p be 'It is cold' and q be 'He wears a hat'. Then using $p \rightarrow q \sqcap \sim p \sqcap q$, we have

It is not cold or he wears a hat.

b) Similarly if we let p be 'Productivity increases' and q be 'Wages rise' then using $p \rightarrow q \sqcap \sim p \sqcap q$ we get

Productivity does not increase or wages rise.

6.2.4.1 Converse, Inverse and Contrapositive

Consider the conditional proposition $p \rightarrow q$. Then the simple propositions $q \rightarrow p$, $\sim p \rightarrow \sim q$ and $\sim q \rightarrow \sim p$ are called respectively, the converse, inverse and contrapositive of $p \rightarrow q$.

Which of these propositions are logically equivalent to $p \rightarrow q$? Consider the truth table below;

p	q	$\sim p$	$\sim q$	Conditional $p \rightarrow q$	Converse $q \rightarrow p$	Inverse $\sim$ $p \rightarrow \sim q$	Contrapositive $\sim q \rightarrow \sim p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

We observe that columns 5 and 8 are identical, thus only the contrapositive $\sim q \rightarrow \sim p$ is logically equivalent to the original conditional proposition $p \rightarrow q$. That is $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

In addition columns 6 and 7 are identical, thus $q \rightarrow p \equiv \sim p \rightarrow \sim q$.

Example 6.11

Give the converse, inverse and contrapositive of each of the following;

- i) If it snows, they will drive the car. ii) Only if Mark studies will he pass the examination.

Solution

- i) Let p be 'It snows' and q be 'They will drive the car'. Then Converse $q \rightarrow p$

If they drive the car, then it will snow.

Inverse $\sim p \rightarrow \sim q$

If it does not snow, then they will not drive the car.

Contrapositive $\sim q \rightarrow \sim p$

If they will not drive the car, then it will not snow.

- i) The given statement can be written as;

If Mark passes the examination, then he studied. Thus if we let p be 'Mark passes the examination' and q be 'Mark studied' then

Converse $q \rightarrow p$

If Mark studies, then he will pass the examination.

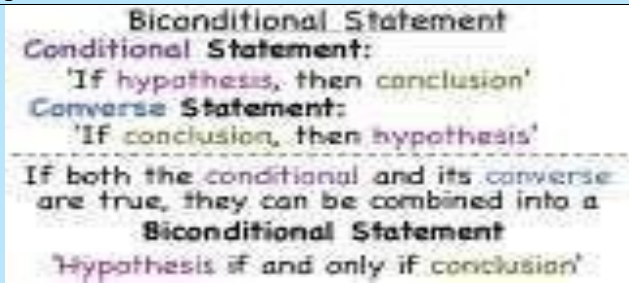
Inverse $\sim p \rightarrow \sim q$

If Mark does not pass the examination, then he did not study.

Contrapositive $\sim q \rightarrow \sim p$

If Mark does not study, then he will not pass the examination.

Etivity 6.2.4: Conditional and Biconditional statements

Numbering and pacing and sequencing	6.2.4
Title	Conditional and biconditional statements
Purpose	To help you to determine truth values of conditional and biconditional statements
Brief summary of overall task	Watch the videos on Logic: Conditional statements by Shaunteaches, Related conditionals by Keith Mann and Examine whether by Doubtnut answer the given questions.
Spark	 Biconditional Statement Conditional Statement: 'If hypothesis, then conclusion' Converse Statement: 'If conclusion, then hypothesis' If both the conditional and its converse are true, they can be combined into a Biconditional Statement 'Hypothesis if and only if conclusion'
Individual contribution	• Watch the three videos • Show that the converse and the inverse are logically equivalent.
Interaction begins	• Post your answers on discussion forum 6.2.4 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	Methods of proof

6.3 Assessment Questions

Show that

$$\begin{aligned} \text{i) } p \sqcap (q \sqcap r) &\sqcap (p \sqcap q) \sqcap (p \sqcap r) \\ \text{ii) } \sim(\sim p) &\sqcap p \\ \text{iii) } p \sqcap q &\sqcap q \sqcap p \end{aligned}$$

6.4 References

1. Goldstein, L., Schneider, D. and Siegel, M., Finite Mathematics and its applications, 7th Ed., Prentice Hall, 1998.
2. Grimaldi, R.P., Discrete and Combinatorial Mathematics. An applied introduction, 5th Ed., Pearson Addison Wesley, 2004.
3. Edgar, G., Goodaire, M., Parmenter, M., Discrete Mathematics with graph theory, 2nd Ed., 2002.
4. Seymour Lipschutz and Marc Lipson. Schaum's Outline Series Discrete Mathematics, 3rd edition, McGraw-Hill Companies, Inc., 2007.
<https://alas.matf.bg.ac.rs/~mi10164/Materijali/DS.pdf>

LECTURER SEVEN**METHODS OF PROOF****7.1 Introduction**

In the previous lesson we considered how to determine the truth values of propositions (statements). In this lesson we move a step further and consider methods of proving mathematical statements. We will consider proof by Induction, Contradiction and Contraposition and direct proof.

7.2 Learning Outcome

By the end of this lesson the learner will be able to:

- i) Prove mathematical statements using direct proof. ii)
- Prove mathematical statements using indirect proofs. iii)
- Prove mathematical statements by induction.

7.2.1 Direct Proof

Direct proof is a way of showing the truth or falsehood of a given statement by a straightforward combination of established facts, usually axioms, existing lemmas and theorems, without making any further assumptions. Logical deduction is employed to reason from assumptions to conclusion.

A direct proof assumes that the hypothesis of a conjecture is true, and then uses a series of logical deductions to prove that the conclusion of the conjecture is true. **Example 7.1**

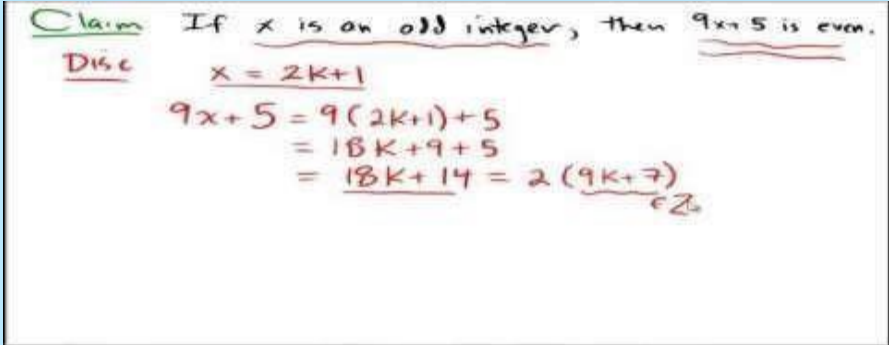
Show that the square of an even integer is even.

Proof

Suppose n is an even integer, then $n = 2m$ for some integer m . Therefore n^2
 $= (2m)^2 = 4m^2 = 2(2m^2)$.

Hence n^2 is even.

Etivity 7.2.1: Direct Proof

Title	Direct Proof
Purpose	To expose you to direct method of proof.
Brief summary of overall task	Watch the video on Direct proof by Centre of Maths and answer the given question.
Spark	
Individual contribution	• Watch the video on direct proof. • Prove that the sum of two odd integers is an even integer.
Interaction begins	• Post your answers on discussion forum 7.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour.
Next	Indirect proof

7.2.2 Indirect Proof

In an indirect proof, instead of showing that the conclusion to be proved is true, you show that all of the alternatives are false. To do this, you must assume the negation of the statement to be proved. Then, deductive reasoning will lead to a contradiction: two statements that cannot both be true.

There are two types of indirect proofs: contradiction, contraposition and counterexample.

7.2.2.1 Proof by Contradiction

Sometimes the only (or the best) way to show why some conclusions holds, is to assume that it doesn't. If this leads to 'nonsense' contrary to reality or a result contrary to the hypotheses, then we have reached a contradiction. Such a proof is called a proof by contradiction.

Example 7.2

Prove that $\sqrt{2}$ is irrational.

p

(A number is rational if it can be written in the form $\frac{p}{q}$ with p and q integers $q \neq 0$ and gcd of p and q is 1 i.e. $(p, q) = 1$ and is irrational if it is not rational.)

Proof

Assume to the contrary that $\sqrt{2}$ is rational. Then p

$\sqrt{2} = \frac{p}{q}$, with $(p, q) = 1$, $q \neq 0$ and p and q are integers.

Therefore $2 = \frac{p^2}{q^2} \Rightarrow p^2 = 2q^2$

Hence p^2 is even so that p is also even. This implies that $p = 2k$ for some integer k , thus $(2k)^2 = 4k^2 = 2q^2$ and therefore $q^2 = 2k^2$.

Thus q^2 and q are also even. Therefore both p and q are even, contradicting our earlier statement. Hence $\sqrt{2}$ is irrational. $\square$

7.2.2.2 Contrapositive Proof

Given a set of hypotheses $H_1, H_2, \dots, H_n$ from which we want to infer a conclusion C , for contrapositive proof we need to show that

$$\sim C \Rightarrow \sim (H_1 \wedge H_2 \wedge \dots \wedge H_n).$$

That is, we show that negation of the conclusion implies negation of the hypothesis.

Example 7.3

Prove that if n^2 is even, then n is even.

Solution

We prove the contraposition that if n is not even, then n^2 is not even

If n is not even (is odd), then $n = 2k + 1$ for some integer k .

Therefore, n^2

$$= (2k + 1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

Hence then n^2 is not even (is odd).

7.2.2.3 Counter Example

A counterexample is a special kind of example that disproves a statement or proposition. Identifying counterexamples is a way to show that a mathematical statement is false.

Obtaining counterexamples is a very important part of mathematics, because doing mathematics requires that you develop a critical attitude toward claims. When you have an idea or when someone tells you something, test the idea by trying examples. If you find a counterexample which shows that the idea is false, that's good. Progress comes not only through doing the right thing, but also by correcting your mistakes.

Example 7.7

- The statement “All prime numbers are odd numbers” is false since for example 2 is a prime number and it's not odd. In this case 2 is counter example.
- The statement “for all real numbers x , we have $(x + 1)^2 = x^2 + 1$ ” is false since for example 1 is a real number and $(1 + 1)^2 \neq 1^2 + 1$ i.e. $4 \neq 2$. In this case $x = 1$ is a counter example.

E-tivity 7.2.2: Indirect Proof

Numbering and pacing and sequencing Title	7.2.2 Indirect proof
Purpose	To expose you to various methods of indirect proof.
Brief summary of overall task	Watch video on Proof by contradiction by GVSUMath, Proof by contradiction by Centre of Maths, Proof by contraposition by GVSUMath and Finding a Counter example by Keith Mann answer the given questions.

Spark	 Proof by Contradiction Theorem: $\sqrt{2}$ is irrational. - Suppose $\sqrt{2}$ was rational - So have n, d integers without common prime factors such that $\sqrt{2} = \frac{n}{d}$ - We will show that n & d are both even. © 2012 Albert R. Meyer February 11, 2012 lec 8P.4
Individual contribution	- Watch the four videos - Prove the following statement by a) contradiction b) contraposition If $3n+5$ is odd, then n is even. - Disproof by counter example that $n^2 + n + 17$ is prime
Interaction begins	- Post your answers on discussion forum 7.2.2 - Read what your colleagues have posted. - In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	- Focussing group discussion - Encouraging lurkers (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This activity should take 1 hour
Next	Proof by Induction

7.2.3 Proof by Induction

This is an important tool used when proving theorems on natural numbers (positive integers).

Theorem 7.1

Let $p(n)$ be a statement that is either true or false for each $n \in N$, then $p(n)$ is true for all $n \in N$ provided that

- $p(1)$ is true.
- For each $k \in N$, if $p(k)$ is true, then $p(k+1)$ is true.

Example 7.4

Prove that $1 + 2 + 3 + \dots + n = \frac{1}{2}n(n+1)$ for any natural numbers n .

Proof

Let $p(n) = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$, then

- i) $RHS = p(1) = \frac{1(1+1)}{2} = 1$ and $LHS = 1$, hence the statement is true for $n = 1$.
- ii) Assume that $p(k)$ is true for some $k \in \mathbb{N}$, that is $p(k) = \frac{k(k+1)}{2}$ and show that $p(k+1)$ is true.

Now,

$$\begin{aligned} p(k+1) &= 1 + 2 + 3 + \dots + k + (k+1) = p(k) + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

Hence $p(k+1)$ is true. Hence by inductive hypothesis the statement is true for all natural numbers n .

Example 7.5

Prove that $7^n - 4^n$ is a multiple of 3 for all n .

Proof

- i) Let $p(n) = 7^n - 4^n$. For $n = 1$, $p(1) = 3$, which is divisible by 3. Hence the statement is true for $n = 1$.
- ii) Next assume that $p(k)$ is true. We want to show that $p(k+1)$ is a multiple of 3.

Now,

$$\begin{aligned}
 p(k+1) &= 7_{k+1} - 4_{k+1} = 7_k(7) - 4_k(4) \\
 &= 7^k(4+3) - 4^k(4) \\
 &= 7^k(4) + 7^k(3) - 4^k(4) \\
 &= 4(7^k - 4^k) + 7^k(3) \\
 &= 4 \square\square p(k) \square\square + 7^k(3) ; \text{ but } p(k) = 3m \text{ for some } m \square N \\
 &= 4(3m) + 7^k(3) \\
 &= 3(4m + 7^k) , \text{ which is a multiple of 3.}
 \end{aligned}$$

Hence $p(k+1)$ is true so that by inductive hypothesis the statement is true for all natural numbers.

Example 7.6

Prove that $2 + 4 + 6 + \dots + 2n = n(n+1)$ for all $n \square\square$.

Solution

Let $p(n) = n(n+1)$. Then

i) If $n = 1$, $LHS = 2$ and $RHS = p(1) = 1(1+1) = 1(2) = 2$. Hence the statement is true for $n = 1$.

ii) Assume that $p(k)$ is true for some $k \square N$, that is $p(k) = k(k+1)$ and show that $p(k+1)$ is true.

Now,

$$p(k+1) = 2 + 4 + 6 + \dots + 2k + 2(k+1) = p(k) + 2(k+1)$$

$$= k(k+1) + 2(k+1)$$

$$= k^2 + k + 2k + 2$$

$$= k^2 + 3k + 2$$

$$= (k+1)(k+2)$$

$$= (k+1) \square \square (k+1) + 1 \square \square$$

Hence $p(k+1)$ is true so that by inductive hypothesis the statement is true for all natural numbers.

E-tivity 7.2.3: Proof by induction

Numbering and pacing and sequencing	7.2.3
Title	Proof by induction
Purpose	To expose you to method of proof by induction
Brief summary of overall task	Watch the video on Proof by induction by Centre of Math on and answer the given question.
Spark	Proof by mathematical induction Let S be a conjecture involving a positive integer n. S is true for all positive integers if the following conditions hold. 1. S is true for $n = 1$ 2. If S is true for $n = k$ 3. Then S is true for $n = k + 1$

Individual contribution	• Watch videos on proof by mathematical • Prove that $1 + 3 + 5 + \dots + (2n - 1) = n^2, \text{ for all } n \in \mathbb{N}$
Interaction begins	• Post your answers on discussion forum 7.2.3 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour.
Next	Introduction to complex numbers

7.3 Assessment

1. Prove that the sum of two odd integers is an even integer.
2. Prove that

$$1^2 + 2^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1), n \in \mathbb{N}.$$

3. Prove that $9^n + 7$ is always divisible by 8.

7.4 References

1. Goldstein, L., Schneider, D. and Siegel, M., Finite Mathematics and its applications, 7th Ed., Prentice Hall, 1998.
2. Grimaldi, R.P., Discrete and Combinatorial Mathematics. An applied introduction, 5th Ed., Pearson Addison Wesley, 2004.

3. Edgar, G., Goodaire, M., Parmenter, M., Discrete Mathematics with graph theory, 2nd Ed., 2002.
4. Seymour Lipschutz and Marc Lipson. Schaum's Outline Series Discrete Mathematics, 3rd edition, McGraw-Hill Companies, Inc., 2007.
<https://alas.matf.bg.ac.rs/~mi10164/Materijali/DS.pdf>

LECTURE EIGHT**COMPLEX NUMBERS****8.1 Introduction**

In this lesson we will study complex numbers. The set of all real numbers $\mathbb{R}$ is incomplete in the sense that standard operations applied to some real numbers do not yield a real number result (e.g., square root: $\sqrt{-1}$). It is surprisingly easy to enlarge the set of real numbers producing a set of numbers that is closed under standard operations, that is the subject of this lecture.

8.2 Learning Outcomes

By the end of this lesson the learner will be able to:

- i) Carry out arithmetic operations on complex numbers
- ii) Determine the modulus and argument of any complex number

iii) State and apply De Moivre's Theorem

8.2.1 Introduction to Complex Numbers

Consider the quadratic equation $x^2 + 1 = 0$. Then $x^2 = -1$, hence $x^2 + 1 = 0$ has no real roots.

To solve an equation of this type, we need another category of numbers called complex numbers.

We define $i = \sqrt{-1}$, where i is an imaginary unit. This can be extended to find the square root of other negative numbers.

$$\begin{aligned} \text{Eg. } \sqrt{-25} &= \sqrt{25 \times -1} \\ &= \sqrt{25} \times \sqrt{-1} \\ &= 5i \end{aligned}$$

Definition 8.1

The general form of a complex number z is $x + iy$, where x and y are real numbers and i is the imaginary unit. The numbers x and y are called the real and imaginary parts of z respectively written $x = \operatorname{Re}(z)$ and $y = \operatorname{Im}(z)$.

For example if $z = 2 + 7i$, then $\operatorname{Re}(z) = 2$ and $\operatorname{Im}(z) = 7$.

The set of all complex numbers is denoted by $\mathbb{C}$.

Definition 8.2

Let $z = x + iy$ be a complex number, then the number $x - iy$ denoted by $\bar{z}$ is called the complex conjugate of z .

Example 8.1

If $z = 3 + 2i$, then $\bar{z} = 3 - 2i$.

8.2.1.1 Arithmetic operations on Complex Numbers

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, then we have

$$1. \quad z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$

$$= (x_1 + x_2) + i(y_1 + y_2)$$

$$2. \quad z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2)$$

$$= (x_1 - x_2) + i(y_1 - y_2)$$

$$3. \quad z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2)$$

$$= x_1 x_2 + x_1 y_2 i + x_2 y_1 i + y_1 y_2 i^2$$

$$= x_1 x_2 + i(x_1 y_2 + x_2 y_1) - y_1 y_2$$

$$= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

In particular

$$\begin{aligned} z_1 z_1 &= (x_1 + iy_1)(x_1 - iy_1) \\ &= x_1^2 - x_1 y_1 i + x_1 y_1 i - y_1^2 i^2 \\ &= x_1^2 + y_1^2 \quad (\text{real number}) \end{aligned}$$

$$4. \quad \frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2}$$

We can rationalize the denominator by multiplying by the conjugate of z_2 in the numerator and denominator;

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} \\ &= \frac{x_1 x_2 - x_1 y_2 i + x_2 y_1 i + y_1 y_2}{x_2^2 - x_2 y_2 i^2 + x_2 y_2 i^2 + y_2^2} \\ &= \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \end{aligned}$$

$$\begin{aligned} &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} . \end{aligned}$$

Example 8.2

Let $z_1 = 1 + i$, $z_2 = 2 + \sqrt{3}i$. Find

a) $z_1 + z_2$ b) $z_1 - z_2$ c) $z_1 z_2$ d) $\frac{z_1}{z_2}$

Solution

$$\begin{aligned} 1. \quad z_1 + z_2 &= (1 + i) + (2 + \sqrt{3}i) \\ &= 3 + (1 + \sqrt{3})i \end{aligned}$$

$$\begin{aligned} 2. \quad z_1 - z_2 &= (1 + i) - (2 + \sqrt{3}i) \\ &= -1 + (1 - \sqrt{3})i \end{aligned}$$

$$\begin{aligned} z_1 z_2 &= (1 + i)(2 + \sqrt{3}i) \\ &= 2 + \sqrt{3}i + 2i + \sqrt{3}i^2 \\ &= 2 - \sqrt{3} + (3 + \sqrt{3})i \end{aligned}$$

4. Rationalizing the denominator, we get

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{1 + i}{2 + \sqrt{3}i} \\ &= \frac{(1 + i)(2 - \sqrt{3}i)}{(2 + \sqrt{3}i)(2 - \sqrt{3}i)} \\ &= \frac{2 - \sqrt{3}i + 2i + \sqrt{3}}{4 - 2\sqrt{3}i + 2\sqrt{3}i + 3} \end{aligned}$$

=

$$\frac{(2 + \sqrt{3}) + (2 - \sqrt{3})i}{7}$$

=

7

$$= \frac{2 + \sqrt{3}}{7} + i \frac{2 - \sqrt{3}}{7}.$$

8.2.1.2 Equal Complex Numbers

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then $z_1 = z_2$ if and only if $x_1 = x_2$ and $y_1 = y_2$.

Example 8.3

Solve for x and y if

a) $3 + 4i = (x + iy)(1 + i)$

b) $3x + 2iy - 7 - 5i = xi - 5y$

c) $\frac{x + iy}{2 + i} = 5 - i$

Solution

a) $3 + 4i = x + ix + iy - y$

$$= (x - y) + i(x + y)$$

Therefore

$$\begin{aligned} x - y &= 3 \\ x + y &= 4 \end{aligned}$$

$$\text{—————}$$

$$-2y = -1$$

$$\square y = \frac{1}{2}$$

$$x = \frac{7}{2}$$

2

b) Rearranging the given equation

$$3x + 5y + i(2y - x) = 7 + 5i$$

Therefore

$$3x + 5y = 7$$

$$-x + 2y = 5 \quad \square \quad x = -1 \quad y = 2$$

$$x + iy$$

c) $\frac{x + iy}{2 + i} = 5 - i$ implies that

$$x + iy = (5 - i)(2 + i)$$

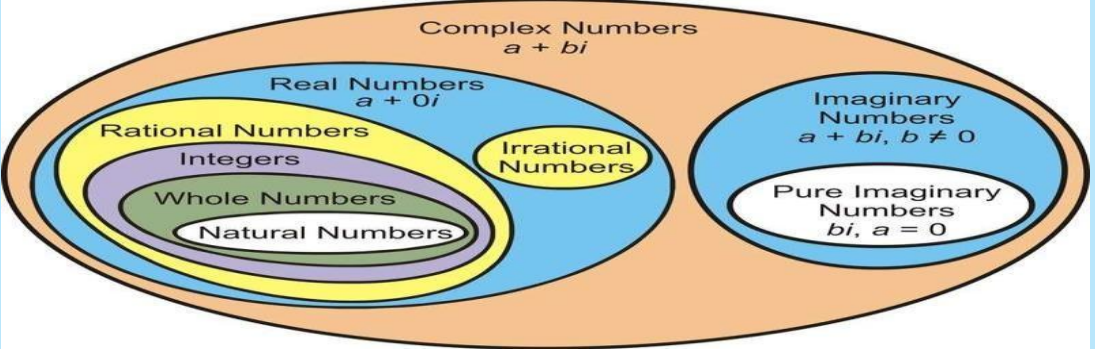
$$= 10 + 5i - 2i + 1$$

$$= 11 + 3i$$

$$\square \quad x = 11 \quad \text{and} \quad y = 3.$$

E-tivity 8.2.1: Introduction to complex numbers

Numbering and pacing and sequencing	8.2.1
Title	Introduction to complex numbers
Purpose	To help you to perform basic operations on complex numbers
Brief summary of overall task	Watch the video on Operations on complex numbers by Professor J and answer the given questions.

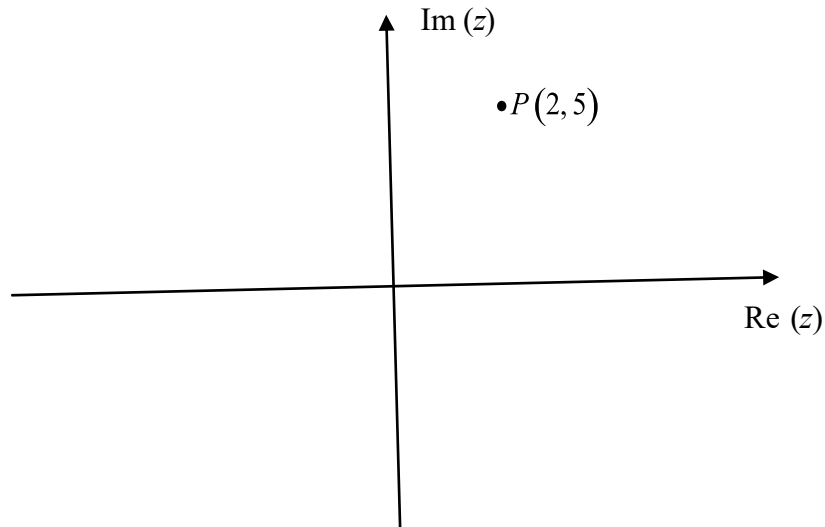
Spark	
Individual contribution	• Watch the video on operations with complex numbers • Answer the questions; 1. Simplify a) $(2 + i)(3 - 4i)$ b) $(5 + 4i)(7 - i)$ c) $(3 - i)(4 - i)$ 2. Express with rational denominators a) $\frac{1 - i}{1 + i}$ b) $\frac{3i - 2}{1 + 2i}$ c) $\frac{5 + 4i}{5 - 4i}$ d) $\frac{1}{(1 + i)^2}$
Interaction begins	• Post your answers on discussion forum 8.2.1 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour.
Next	Geometric representation of complex numbers

8.2.2 Geometric Representation of Complex numbers

The complex number $z = x + iy$ can be represented by the ordered pair (x, y) . This suggests that z can be represented in the xy -plane called the z -plane or the complex plane or the Argand diagram. On an Argand diagram, real numbers are represented on the x -axis and imaginary numbers on the y -axis.

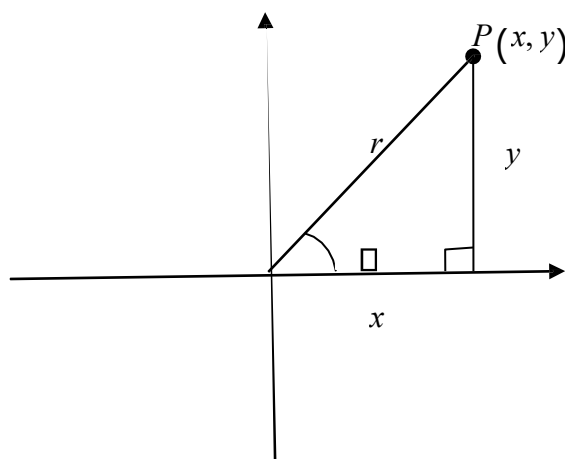
Thus the complex number P in the figure below for example can be read as $(2, 5)$ or $2 + 5i$.

The Argand Diagram



8.2.2.1 Modulus and Argument

Let P represent the complex number z where $z = x + iy$, then P has Cartesian coordinates (x, y) . Consider the figure below



By Pythagoras Theorem,

$$r^2 = x^2 + y^2 \quad \square \quad r = \sqrt{x^2 + y^2}.$$

Also

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r} \quad \text{and} \quad \tan \theta = \frac{y}{x}.$$

Definition 8.3

Given any complex number $z = x + iy$, the modulus of z , denoted $|z|$ is defined by

$$|z| = r = \sqrt{x^2 + y^2}.$$

The argument of z denoted $\arg z$, is defined by

$$\arg z = \theta = \tan^{-1} \frac{y}{x}.$$

There is an infinite set of angles whose tangent is $\frac{y}{x}$, so there is also an infinite set of arguments

for $x + iy$. But the position of (x, y) is unique and corresponds to only one value of θ in the range $-\pi < \theta \leq \pi$. This value is called the principal argument of z .

Now,

$$\cos \theta = \frac{x}{r} \quad \Rightarrow \quad x = r \cos \theta$$

and

$$\sin \theta = \frac{y}{r} \quad \Rightarrow \quad y = r \sin \theta$$

Thus a non-zero complex number z , can also be written as

$$z = x + iy = r \cos \theta + i r \sin \theta$$

$$= r (\cos \theta + i \sin \theta) \quad \dots\dots\dots (\theta)$$

(r, θ) is called the polar form of the complex number z . The polar coordinates of z are (r, θ) .

(where θ is the principal value of $\arg z$).

Example 8.4

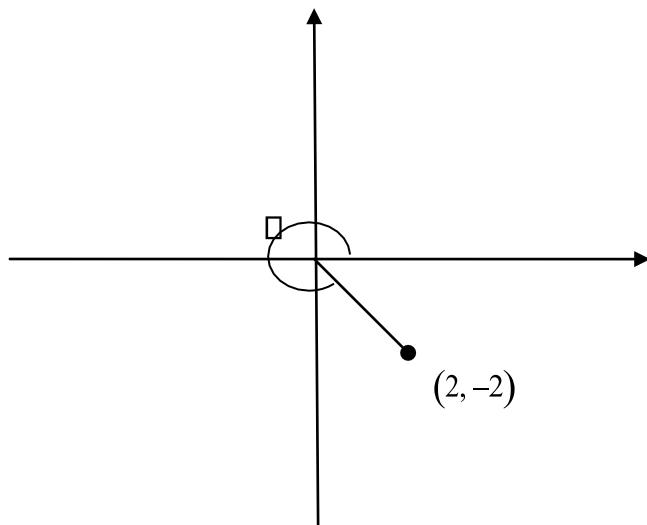
Let $z = 2 - 2i$, find $|z|$, $\arg z$ and the polar form of z .

Solution

Modulus

$$\begin{aligned} |z| &= \sqrt{2^2 + (-2)^2} \\ &= 2\sqrt{2} \end{aligned}$$

Also z is in the 4th quadrant;



$$\begin{aligned} \text{Principal value} &= \tan^{-1} \frac{-2}{2} = \tan^{-1} (-1) = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4} \end{aligned}$$

Therefore $\arg z = \frac{7}{4}\pi + 2n\pi$, n

$= 0, 1, 2, \dots$

And the polar form of z is

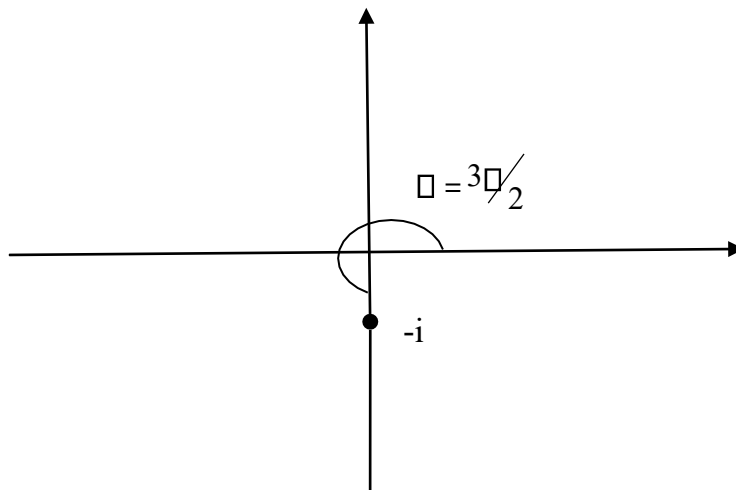
$$z = 2\sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right).$$

Example 8.5

Let $z = -i$, find $|z|$ and $\arg z$

Solution

The complex number $z = -i$ is on the y -axis;



$$z = 0 + i(-1) \text{ and } r = |z| = \sqrt{0^2 + (-1)^2} = 1$$

Principal value is $\theta = \frac{3\pi}{2}$.

$$\theta \arg z = \frac{3\pi}{2} + 2n\pi, n = 0, 1, 2, \dots$$

8.2.2.2 Products and Quotients of Complex Numbers

We can use polar form to multiply complex numbers.

If $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$, then

$$z_1 z_2 = r_1 (\cos \theta_1 + i \sin \theta_1) \cdot r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$= r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2] + i (\cos \theta_2 \sin \theta_1 + \cos \theta_1 \sin \theta_2)$$

$$= r_1 r_2 [\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)]$$

Thus $(\theta_1 + \theta_2)$ is the principal argument of $z_1 z_2$, hence

$$\arg (z_1 z_2) = \arg z_1 + \arg z_2.$$

Also

$$|z_1 z_2| = |z_1| |z_2|.$$

Since multiplication and division are inverse operations, it can be deduced that z_1

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \frac{[\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)]}{1}$$

$$z_2 = r_2 [\cos \theta_2 + i \sin \theta_2]$$

$$\arg z_1$$

$$\arg \frac{z_1}{z_2} = \arg z_1 - \arg z_2$$

and

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}.$$

Remark 8.1

It is sometimes convenient to write $\cos \theta + i \sin \theta$ as $e^{i\theta}$ or *cis* θ .

so that

$$r \text{cis} \theta = r e^{i\theta} = r (\cos \theta + i \sin \theta).$$

The expression $e^{i\theta} = \cos \theta + i \sin \theta$ is called the Euler's formula. The Euler's formula can be generalized to $e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y)$.

Example 8.6

If $z_1 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $z_2 = 1 + i\sqrt{3}$, find

a) $|z_1 z_2|$

b) $\arg(z_1 z_2)$

c) $\arg$
 $\frac{z_1}{z_2}$

Solution

a) $|z_1 z_2| = \left| -\frac{1}{2} + i\frac{\sqrt{3}}{2} \right| \left| 1 + i\sqrt{3} \right|$

$$= 1 \cdot 2 = 2.$$

b) $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

Now,

$$\begin{aligned}\arg z &= \tan^{-1} \frac{-\sqrt{3}}{1} = \tan^{-1} \left(-\frac{\sqrt{3}}{1} \right) \\ &= -\frac{\pi}{3} \\ &= -\frac{2\pi}{3}.\end{aligned}$$

and

$$\arg z_2 = \tan^{-1} \left(\frac{\sqrt{3}}{3} \right) = \frac{\pi}{6}.$$

Thus

$$\arg(z_1 z_2) = -\frac{2\pi}{3} + \frac{\pi}{6} = -\frac{\pi}{2}.$$

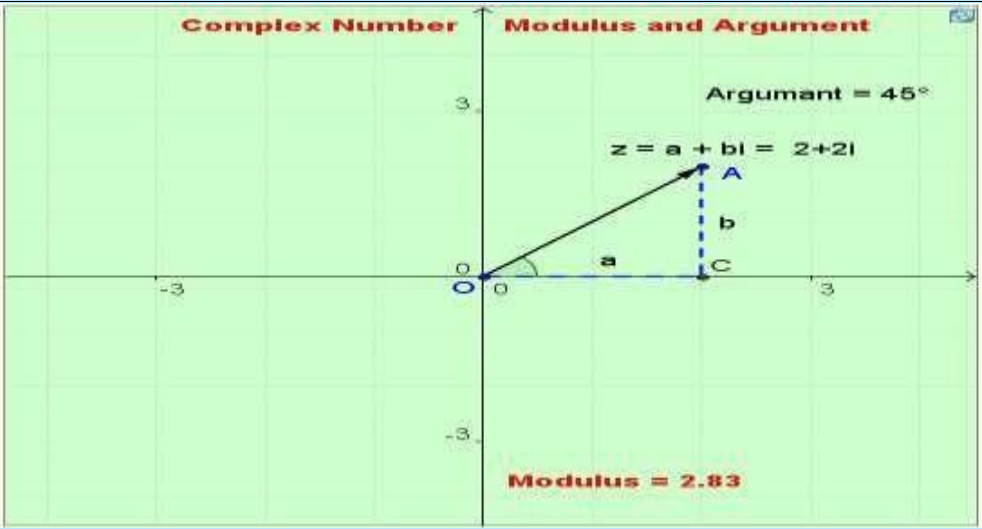
c)

$$\arg z_1 = -\frac{\pi}{3} = -\frac{2\pi}{6}.$$

$$\arg z_2 = \frac{\pi}{6}$$

E-tivity 8.2.2: Geometric representation of complex numbers

Numbering and pacing and sequencing Title	8.2.2 Geometric representation of complex numbers
Purpose	To help you to represent any complex number in polar form

Brief summary of overall task	Watch the video on Polar form of complex numbers by Vasudeva Vamadevan and answer the given questions.
Spark	
Individual contribution	• Watch the video on Polar form of complex numbers • Answer the questions; 1. Find the modulus and argument of w given that $w = \frac{3-i}{2-i}$. 2. Express the quotient $\frac{1+i}{3-i}$ in its polar form.
Interaction begins	• Post your answers on discussion forum 8.2.2 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	De Moivre's Theorem and its applications

8.2.3 De Moivre's Theorem and its Applications 8.2.3.1

Powers of Complex Numbers

Let $z = r(\cos \theta + i \sin \theta)$ and let $z_1 = z_2 = z$, then

$$z_1 z_2 = z^2 = r^2 (\cos 2\theta + i \sin 2\theta) .$$

In general, if $z_1 = z_2 = \dots = z_n = z$, then $z^n = r^n (\cos n\theta + i \sin n\theta) \dots\dots\dots 1) .$

If $r = 1$, then equation 1) becomes

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta .$$

Theorem 8.1 (De Moivre's Theorem) Let

θ be any angle and n be any integer, then

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta .$$

If n is a negative integer, then

$$(\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta) .$$

Proof

We wish to show that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$. We proceed by induction. If $n = 1$,

$$LHS = (\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta \text{ and } RHS = \cos 1(\theta) + i \sin 1(\theta) = \cos \theta + i \sin \theta .$$

Hence the result is true for $n = 1$.

Assume that the result holds for some integer k , that is $(\cos \theta$

$$+ i \sin \theta)^k = \cos k\theta + i \sin k\theta .$$

Then

$$\begin{aligned} (\cos \theta + i \sin \theta)^{k+1} &= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta) \\ &= (\cos k\theta + i \sin k\theta) (\cos \theta + i \sin \theta) \end{aligned}$$

$$\begin{aligned}
 &= \cos k\theta \cos \theta + i \cos k\theta \sin \theta + i \sin k\theta \cos \theta - \sin k\theta \sin \theta \\
 &= \cos k\theta \cos \theta - \sin k\theta \sin \theta + i (\cos k\theta \sin \theta + \sin k\theta \cos \theta) \\
 &= \cos(k\theta + \theta) + i \sin(k\theta + \theta) \\
 &= \cos(k+1)\theta + i \sin(k+1)\theta
 \end{aligned}$$

Hence the theorem is true for the positive integer $k+1$. Hence by the principle of mathematical induction it holds for any positive integer. It can similarly be shown that

$$(\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta). \quad \square$$

Example 8.7

Use De Moivre's theorem to evaluate

$$\begin{aligned}
 \text{a) } & (4 + i4\sqrt{3})^5 \quad \text{b) } \frac{1 + i\sqrt{3}}{1 - i\sqrt{3}}^{10}
 \end{aligned}$$

Solution

$$\sqrt{} \quad /$$

We write $4 + i4\sqrt{3}$ in polar form;

$$|4 + i4\sqrt{3}| = \sqrt{4^2 + (4\sqrt{3})^2} = 8 \quad \text{and} \quad \arg(4 + i4\sqrt{3}) = \theta$$

$$4 + i4\sqrt{3} = 8 \left(\cos \frac{\theta}{3} + i \sin \frac{\theta}{3} \right)$$

By De Moivre's Theorem,

$$\begin{aligned}
 (4 + i)^{3/5} &= 8^{3/5} (\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}) \\
 &= 8^{3/5} (\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}) \\
 &= 8^{3/5} (\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5})
 \end{aligned}$$

Example 8.8

By using De Moivre's theorem show that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

Solution

$$\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta}$$

By De Moivre's theorem,

$$\begin{aligned}
 \cos 3\theta + i \sin 3\theta &= (\cos \theta + i \sin \theta)^3 \\
 &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \\
 &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta)
 \end{aligned}$$

Equating the real and imaginary parts,

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta \dots\dots\dots i)$$

$$\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta \dots\dots\dots ii)$$

$$\tan 3\theta = \frac{3 \cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3 \cos \theta \sin^2 \theta}$$

$$\begin{aligned}
 & \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\cos \theta \sin^2 \theta} \\
 &= \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\cos \theta \sin^2 \theta} \\
 &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} .
 \end{aligned}$$

Example 8.9

Use De Moivre's theorem to show that

$$\begin{aligned}
 \frac{\sin 5\theta}{\sin \theta} &= 16 \cos^4 \theta - 12 \cos^2 \theta + 1
 \end{aligned}$$

Solution

By De Moivre's theorem,

$$\cos 5\theta + i \sin 5\theta = (\cos \theta + i \sin \theta)^5$$

But

$$\begin{aligned}
 (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \\
 &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i(5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta)
 \end{aligned}$$

Equating the real and imaginary parts

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \text{ and } \sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

Now,

$$\frac{\sin 5\theta}{\sin \theta} = 5\cos^4 \theta - 10 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\begin{aligned} &= 5\cos^4 \theta - 10 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 \\ &= 5\cos^4 \theta - 10 \cos^2 \theta + 10 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta \\ &= 16 \cos^4 \theta - 12 \cos^2 \theta + 1. \end{aligned}$$

8.2.3.2 Roots of Complex Numbers

A complex number z is said to be the n^{th} root of the complex number z_0 if

$$z^n = z_0 \dots\dots\dots 1)$$

Thus if $z = r (\cos \theta + i \sin \theta)$ and $z_0 = r_0 (\cos \theta_0 + i \sin \theta_0)$, then we have

$$r^n (\cos n\theta + i \sin n\theta) = r_0^n (\cos \theta_0 + i \sin \theta_0)$$

Since $|\cos \theta + i \sin \theta| = 1$ for all real θ , then

$$r^n = r_0^n \quad \text{and} \quad r = r_0^{1/n}$$

and

$$n\theta = \theta_0 + 2\pi k, \quad \text{for } k = 0, 1, 2, \dots$$

$$\theta = \frac{\theta_0 + 2\pi k}{n}, \quad \text{for } k = 0, 1, 2, \dots, n-1$$

$z = r_0^{1/n} [\cos \frac{\theta_0 + 2\pi k}{n} + i \sin \frac{\theta_0 + 2\pi k}{n}]$ is the n^{th} root of z_0 for $n = 0, 1, 2, \dots, n-1$.

$$\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

Thus there are n distinct roots to be determined, the rest are merely repeated.

Example 8.10

Find all values of z for which $z^5 + 32 = 0$.

Solution

$$z^5 + 32 = 0 \Rightarrow z^5 = -32$$

In polar form

$$-32 = 32 \cos(\pi + 2\pi k) + i \sin(\pi + 2\pi k), \text{ where } k = 0, 1, 2, \dots$$

Now,

$$z = (-32)^{1/5} = 32^{1/5} \cos(\pi/5 + 2\pi k) + i \sin(\pi/5 + 2\pi k)$$

$$= 2 \cos(\pi/5 + 2\pi k) + i \sin(\pi/5 + 2\pi k), \text{ where } k = 0, 1, 2, 3, 4.$$

The values of z are;

$$w = 2 \cos(\pi/5) + i \sin(\pi/5), \text{ for } k = 0$$

$$w_1 = 2 \cos(3\pi/5) + i \sin(3\pi/5), \text{ for } k = 1$$

$$w_2 = 2(\cos\pi + i \sin\pi), \text{ for } k = 2$$

$$w = 2 \cos(7\pi/5) + i \sin(7\pi/5)$$

$$z_k = 2 \cos \frac{2\pi k}{5} + i \sin \frac{2\pi k}{5}, \text{ for } k = 3$$

$$w = 2 \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}, \text{ for } k = 4$$

Example 8.11

Find all the cube roots of $-\sqrt{2} + i\sqrt{2}$.

Solution

Let $z_0 = -\sqrt{2} + i\sqrt{2}$. Then

$$|z_0| = \sqrt{2+2} = 2 \text{ and } \arg z_0 = \tan^{-1}(-1) = 34^\circ. \text{ Thus}$$

$$z_0 = 2 \cos (34^\circ + 2\pi k) + i \sin (34^\circ + 2\pi k), k = 0, 1, 2, \dots$$

Now,

$$z^3 = z_0^3 = 2^3 \cos (3 \times 34^\circ + 2\pi k) + i \sin (3 \times 34^\circ + 2\pi k), k = 0, 1, 2$$

$$= 8 \cos 102^\circ + i \sin 102^\circ, k = 0, 1, 2.$$

$$= 8 \cos 102^\circ + i \sin 102^\circ$$

The three roots are

$$w = 2 \cos \frac{102^\circ}{3} + i \sin \frac{102^\circ}{3}, \text{ for } k = 0$$

$$w = 2^{1/3} \cos \frac{2\pi k}{3} + i \sin \frac{2\pi k}{3}, \quad \text{for } k = 1$$

$$w = 2^{1/3} \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}, \quad \text{for } k = 2.$$

Example 8.12

1^6

Evaluate .

$$\frac{1+i}{1-i}$$

$1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$ and $1-i = \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)$ we have

$$\begin{aligned} \sqrt[6]{\frac{1+i}{1-i}} &= \left\{ \frac{\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)}{\sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)} \right\}^{\frac{1}{6}} \\ &= \left\{ \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)^{-1} \right\}^{\frac{1}{6}} \\ &= \left\{ \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right\}^{\frac{1}{6}} \\ &= \left\{ \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^2 \right\}^{\frac{1}{6}} \\ &= \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^{\frac{1}{3}} \\ &= \cos \left(\frac{\pi}{12} + \frac{2\pi k}{3} \right) + i \sin \left(\frac{\pi}{12} + \frac{2\pi k}{3} \right), \quad k = 0, 1, 2. \end{aligned}$$

The three roots are;

Solution

Since

$$\sqrt[6]{1+i}$$

$$\sqrt[6]{1-i}$$

$$\sqrt[6]{1-i}$$

$$w_0 = \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}, \quad \text{for } k = 0$$

$$w_1 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}, \quad \text{for } k = 1$$

$$w_2 = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}, \quad \text{for } k = 2.$$

E-tivity 8.2.3: De Moivre's Theorem and its applications

Numbering and pacing and sequencing Title	8.2.3 De Moivre's Theorem and its applications
Purpose	To help you to apply De Moivre's Theorem to simplify trigonometric identities and find roots of complex numbers
Brief summary of overall task	Watch videos on Using De Moivre's Theorem by Mattam66, De Moivre's Theorem by Tayler Wallance and Problems on De Moivre's Theorem by MA Classes and answer the given questions.
Spark	De Moivre's Theorem if $r(\cos \theta + i \sin \theta)$ is a complex number, then $[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$ or $[rcis\theta]^n = r^n [cisn\theta]$

Individual contribution	• Watch the three videos • Use De Moivre's Theorem to; a) Simplify $\frac{\cos 3\theta + i \sin 3\theta}{(\cos \theta + i \sin \theta)^2}$ i) $(\cos \theta + i \sin \theta)^2 (\cos 3\theta + i \sin 3\theta)$ ii) $(\cos \theta + i \sin \theta)^2$ b) Find $\left(3 - i \right)^{\frac{1}{6}}$.
Interaction begins	• Post your answers on discussion forum 8.2.3 • Read what your colleagues have posted. • In a sentence or two, comment on what two of your colleagues have posted keeping netiquette in mind
E-moderator interventions	• Focussing group discussion • Encouraging lurkers (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This activity should take 1 hour
Next	End of the lessons

8.3 Assessment

1. Solve for x and y

$$a) \frac{2 + 5i}{1 - i} = x + iy \quad b) 2x - 3iy + 4ix - 2y - 5 + 10i = x + y + 2 - (y - x + 3)i$$

2. Use De Moivre's theorem to show that

$$\tan^3 \theta - 5 \tan \theta = 10 \tan^3 \theta - \tan^5 \theta$$

$$4 \tan^4 \theta - 4 \tan^2 \theta = 0$$

$$i) \tan 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad ii) \tan 5\theta = \frac{5 \tan \theta (1 - 10 \tan^2 \theta + 5 \tan^4 \theta)}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$iii) \cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

3. Solve the equation $z^4 + 16i = 0$.

8.4 References

1. Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
2. Spiegel, M. (1999). Shaum's Outline of Theory and Problems of Complex Variables, McGraw-Hill.

ANSWERS

Lesson One Assessment Questions

1. a) $2\frac{5}{4}$ b) $\frac{3}{4}$ c) $-\frac{5}{2}$

4. a) Either the set of all x such that $x \leq -\frac{2}{3}$ or the set of all x such that $x \geq \frac{3}{4}$.

b) Either the set of all x such that $-3 \leq x \leq 5$.

Lesson Two Assessment Questions

1. $-\frac{7}{7}$ a) -7 b) $\frac{27}{27}$

2. $a = 4$ $b = -13$ $4x^3 - 13x - 6 = (x + 2)(2x - 1)(2x - 3)$

Lesson Three Assessment Questions

1. 1820

2. a) 144 b) 7776 c) 756

Lesson Four Assessment Questions

1. $1 + 12x + 54x^2 + 100x^3 + \dots$

2. $(1 + 8x)^{1/2} \approx 1 + 4x - 8x^2 + 32x^3$ and $\sqrt{3} = 1.73205$

Lesson Five Assessment Questions

1. a) 12 b) 54 c) 7

Lesson Eight Assessment Questions

1. a) $x = -\frac{3}{2}$ and $y = \frac{7}{2}$ b) $x = -\frac{53}{7}$ and $y = -\frac{34}{7}$

3. $w_0 = 2\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

$w_1 = 2\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$

$w_2 = 2\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$

$w_3 = 2\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$

RESOURCES

1. Backhouse, J. and Houldsworth, S. (1985). Pure Mathematics Book I, Longman Publishers Ltd.
2. Bostock, L., Chandler, S. and Rourke, C. (1982). Further Pure Mathematics. Stanley Thornes Publishers Ltd.
3. Larson, R. and Hostetler, R. (1997). Precalculus, Houghton Company.
4. Goldstein, L., Schneider, D. and Siegel, M., Finite Mathematics and its applications, 7th Ed., Prentice Hall, 1998.
5. Grimaldi, R.P., Discrete and Combinatorial Mathematics. An applied introduction, 5th Ed., Pearson Addison Wesley, 2004.
6. Edgar, G., Goodaire, M., Parmenter, M., Discrete Mathematics with graph theory, 2nd Ed., 2002.
7. Seymour Lipschutz and Marc Lipson. Schaum's Outline Series Discrete Mathematics, 3rd edition, McGraw-Hill Companies, Inc., 2007.
<https://alas.matf.bg.ac.rs/~mi10164/Materijali/DS.pdf>